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Scale Sensitivity of Regional Uncertainty in Physics-Informed Neural Networks: A Full-Network Gauss–Newton Study

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Abstract

Regional uncertainty should identify where a physics-informed neural network is inaccurate. We assess full-network Gauss–Newton covariances under a homoscedastic working likelihood on training-weighted residuals, using 17 admitted Burgers and Allen–Cahn runs with two architectures. Uncertainty allocation depends strongly on prior and noise scales. Across eight Burgers–MLP seeds, five refined contrast-matching crossings span more than four orders of magnitude in effective precision; three other seeds retain a deficit at every sampled setting. The conditional residual-noise maximum-likelihood setting under-covers both regions around all eight trained predictions. A prior centred on the fitted checkpoint preserves the baseline deficit in all 17 runs; affine accuracy improvements at the noise-MLE setting do not reliably calibrate both regions. At the five trained-centre contrast-matching roots, a reference-informed oracle targeting 68% bulk coverage gives only 30.3–55.7% sharp one-standard-deviation coverage. Reference-informed last-layer fits leave below 4% of the original sharp-region error energy on held-out time blocks in $16/17$ runs, demonstrating feature expressivity. Spectral weighting explains scale-dependent allocation; Lean verifies conditional limits and bounds on contrast, changes of predictive mean and coverage. Regional RMS agreement is insufficient for calibrated intervals: allocation and coverage require separate assessment.

Keywords: physics-informed neural networks, uncertainty quantification, Gauss–Newton covariance, calibration, formal verification

1 Introduction

Physics-informed neural networks (PINNs) approximate PDE solutions by minimising residuals of the governing equations, initial and boundary conditions, and available observations (Raissi et al., 2019). Automatic differentiation supplies the required derivatives. The resulting field is continuous, but training does not ensure accuracy: imbalanced loss gradients, differential-operator conditioning and sampling can impede convergence (Wang et al., 2021; Rathore et al., 2024; Daw et al., 2023). For thin layers or interfaces, the remaining error can be concentrated in a small region. Domain-averaged accuracy and coverage can conceal that concentration.

Scientific-machine-learning uncertainty methods target observation noise, limited information about model parameters, or error in an approximate numerical solution (Psaros et al., 2023). These sources need not have the same spatial pattern. Bayesian PINNs combine physical constraints with

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noisy observations in a probabilistic model, propagating parameter uncertainty to predictions of the solution (Yang et al., 2021). Deep ensembles combine independently trained predictors, so variation between fitted solutions contributes to their predictive uncertainty (Lakshminarayanan et al., 2017).

Local Gaussian methods provide a less costly alternative to posterior sampling by approximating parameter uncertainty using curvature around a trained network (Daxberger et al., 2021). Local linearisation then propagates that parameter covariance through the output tangents to obtain predictive variances (Immer et al., 2021b). Marginal-likelihood methods use the training data to select hyperparameters such as the prior precision (Immer et al., 2021a). The linearisation point need not optimise the resulting affine model, however, and this distinction affects evidence-based selection (Antorán et al., 2022). For PINNs, HypoPINN uses a diagonal-Fisher approximation to quantify predictive uncertainty in hypocentre localisation (Izzatullah et al., 2022). Full-Hessian Laplace inference also permits evidence-based selection of prior strength and the relative weights of PDE, boundary and observation terms (Graczyk and Witkowski, 2026).

Error-bound and calibration methods connect uncertainty to different accuracy criteria. PDE stability estimates can relate PINN solution error to training residuals under stated assumptions (Mishra and Molinaro, 2023). Available solution-error bounds can specify spatially varying likelihood widths in a second Bayesian fitting stage (Flores et al., 2025). Physics-informed conformal prediction uses PDE residuals as nonconformity scores to calibrate coverage in residual space without labelled solutions (Gopakumar et al., 2025). Conformal calibration of PINN predictions instead uses labelled solution values in a calibration set, with local quantile estimation allowing spatially adaptive bands (Yu et al., 2026). These approaches motivate separating uncertainty from numerical accuracy: a covariance derived from a training likelihood still needs to be checked against the error in the computed solution, particularly where that error is concentrated.

We examine whether curvature-based uncertainty identifies numerical errors near sharp layers in solutions of the Burgers and Allen–Cahn equations. We study uncertainty in the solution around a fixed trained network, without additional output noise. For output tangent $g(x) = \nabla_\theta u_\theta(x)$ and the training-weighted residual Gauss–Newton matrix F, the unit-noise baseline is

$$\sigma^2(x) = g(x)^\top \Sigma g(x), \qquad \Sigma = (\alpha I + F)^{-1}.$$

The covariance uses every network parameter. We compare its sharp-to-bulk contrast with each predictive centre's error contrast: the trained output, affine posterior means under zero-mean and θ-centred priors, and a nonlinear shifted-network diagnostic. Reference solutions are used for evaluation, not covariance construction.

Under this training-weighted homoscedastic likelihood, the 17 admitted Burgers and Allen–Cahn runs show a scale-sensitive baseline mismatch (Table 2). Figure 1 illustrates the regional mismatch in one run. The baseline deficit persists under the θ-centred prior in all 17 runs. At weaker precision its affine mean can improve accuracy without calibrating both regions, and the improvement need not transfer to the shifted nonlinear network. A reference-informed oracle matches regional RMS contrasts and calibrates bulk widths, yet sharp intervals under-cover at all five observed trained-centre crossings. Readout refits test feature expressivity; spectral analysis explains covariance weighting.

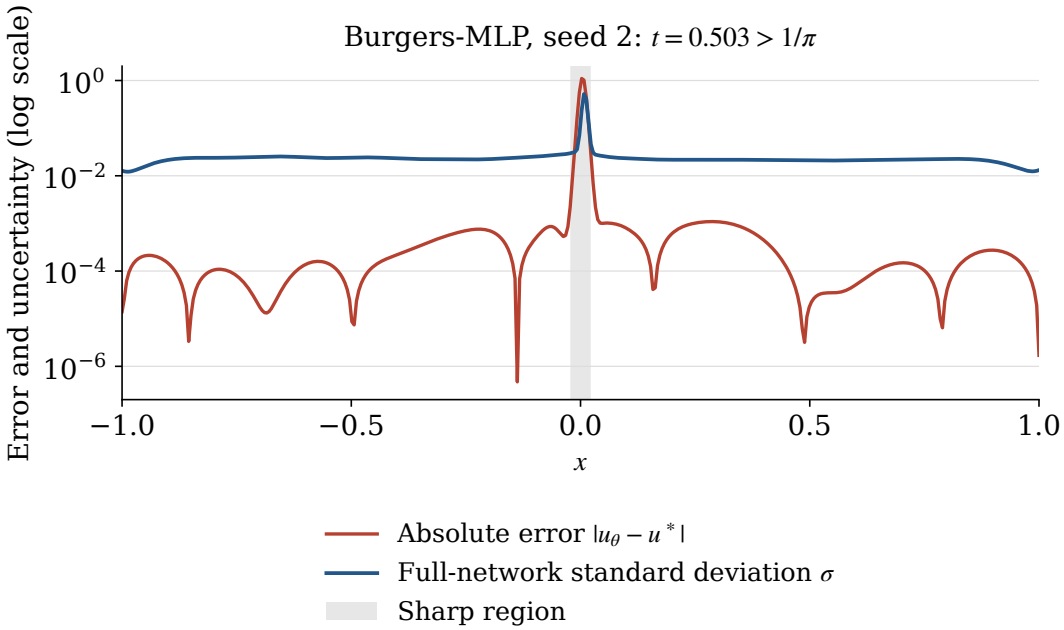


Figure 1: Burgers–MLP seed 2: absolute error and full-network baseline standard deviation at $t = 0.503$, the stored slice nearest $t = 0.5$, after breaking time $1/\pi$. The logarithmic scale is shared without rescaling; shading marks the global sharp region of Definition 1. Error is more concentrated near the layer. Here $|e| > \sigma$ near its centre; this is seed-specific, with post-formation calibration varying across seeds (Table 3). Full-grid contrasts are in Table 2.

Contributions.

- *Scale sensitivity and regional coverage (Section 3.1).* The tested scales and centres do not reliably calibrate both regions. The baseline deficit survives the alternative prior centre, and direct root fields confirm that matching regional RMS contrasts is insufficient after bulk calibration.
- *Reference feature expressivity (Section 3.2).* Using reference values to refit the regional readout leaves below 4% of the trained sharp-region error energy on held-out time blocks in $16/17$ runs. This tests feature expressivity independently of residual-only coefficient selection.
- *Spectral weighting (Section 3.3).* The modal quotient explains scale dependence; two-subspace bounds constrain contrast but are too wide to certify the measured baseline deficit. Lean verifies the spectral bounds and conditional limits, prior-centre bounds and coverage envelopes; reported run counts remain numerical.

Empirical scope. All eight attempted Burgers–MLP seeds are admitted. Each other nonlinear family has three admitted runs selected by the error criterion in Section 2 and is exploratory. Attempted counts and rejected settings remain explicit. The results concern the specified raw residual metric and reference-based partition.

2 Setting and Definitions

We specify the equations, admission rule and frozen affine model before comparing regional uncertainty under changes of scale and predictive centre.

Equations and trained networks. On $(x,t) \in [-1,1] \times [0,1]$, the viscous Burgers equation is

$$u_t + uu_x = \nu u_{xx}, \qquad \nu = 0.01/\pi, \qquad u(x,0) = -\sin(\pi x), \qquad u(\pm 1, t) = 0.$$

The positive-viscosity solution remains smooth and develops a thin layer near $x = 0$. We use the inviscid breaking time $1/\pi$ to define pre- and post-formation windows. The Allen–Cahn equation is

$$u_t = \varepsilon^2 u_{xx} + 5(u - u^3), \qquad \varepsilon^2 = 10^{-4}, \qquad u(x,0) = x^2 \cos(\pi x),$$

with periodic boundaries and interfaces of width $O(\varepsilon)$ between phases $u \approx \pm 1$. A contextual linear control is recorded in Appendix A; it is outside the nonlinear scale study.

The independent-feature MLP has 20 spatial and 20 temporal sub-networks feeding a fusion head, with $P = 196{,}009$ parameters. Replacing its input layer by random Fourier features gives $P = 275{,}369$ at bandwidth FS (Tancik et al., 2020). Both are trained in float64 by Adam (Kingma and Ba, 2015) followed by Levenberg–Marquardt (Levenberg, 1944; Marquardt, 1963). Evaluation uses a uniform 400×200 grid (80,000 points); the residual Jacobian has $N = 6400$ rows: 5000 collocation, 1000 initial-condition and 400 boundary rows. The Burgers reference uses the Cole–Hopf solution with 80-node Gauss–Hermite quadrature, agreeing with independent log-space quadrature to 4.5×10^{-14}. The Allen–Cahn reference uses fourth-order exponential time differencing with 512 modes and 1000 steps (Kassam and Trefethen, 2005); refinement to 2048 modes changes the canonical contrast by less than 0.1%. Appendix A records the training details. Unless a seed count is stated, numerical illustrations use Burgers–MLP seed 2 or Allen–Cahn seed 0.

Definition 1 (Region partition) *Let p_{70}, p_{95} be the 70th and 95th percentiles of $|u_x^*|$ over the whole evaluation grid. Define*

$$\mathcal{S} = \{|u_x^*| \geq p_{95}\}, \qquad \mathcal{B} = \{|u_x^*| < p_{70}\}, \qquad \mathcal{T} = (\mathcal{S} \cup \mathcal{B})^c.$$

These are the sharp, bulk and transitional regions.

The reference partition is used only for evaluation. It follows moving and narrowing features: the Allen–Cahn interface moves by 58 grid cells, and sharp Burgers cells per slice fall from 58 at $t = 0.13$ to 8 at $t = 0.5$. No sampled reference-gradient magnitude equals either threshold. The top-5% definition alone does not establish resolution (Appendix A). For Burgers, the sharp threshold is 3.34, against maximum 116.8, and 69% of sharp points precede formation; temporal results therefore also use the two windows separately.

Definition 2 (Run admission) *A trained prediction is admitted if its relative error outside the sharp region satisfies*

$$\frac{\|(u_\theta - u^*)|_{\mathcal{B}\cup\mathcal{T}}\|_2}{\|u^*|_{\mathcal{B}\cup\mathcal{T}}\|_2} < \tau, \qquad \tau = 0.03 \text{ for Allen–Cahn}, \quad 0.02 \text{ otherwise}.$$

These trained-network decisions remain fixed in every scale and centre comparison. Excluding $\mathcal{S}$ avoids selecting directly on sharp error, but selecting a small bulk error can enlarge its regional ratio.

configuration	admitted/attempted	error range	median	τ
Fisher–KPP	$0/5$	81.7–170.5%	100.0%	2%
Nagumo, both initial conditions	$0/10$	75.9–90.3%	80.5%	2%
Kuramoto–Sivashinsky, $T = 100$	$0/5$	98.2–103.8%	100.4%	2%
Kuramoto–Sivashinsky, $T = 10$	$0/5$	9.9–17.0%	12.8%	2%
Burgers, MLP	$8/8$	0.05–0.40%	0.08%	2%
Burgers, Fourier	$3/8$	0.37–1.48%	0.78%	2%
Allen–Cahn, MLP	$3/5$	1.39–2.33%	2.33%	3%
Allen–Cahn, Fourier	$3/4$	1.64–2.60%	2.34%	3%
Convection–diffusion, MLP control	$3/4$	0.19–0.31%	0.21%	2%

Table 1: Admission accounting and relative L^2 error on $\mathcal{B} \cup \mathcal{T}$. Rejected rows include every attempted run; admitted rows include admitted runs only. Nagumo pools five seeds for each of two initial conditions. The control varies paired pulse width and diffusivity at one seed. Its narrowest MLP run is rejected at 62.2%; a separate Fourier attempt is rejected at 63.3%. All Kuramoto–Sivashinsky attempts also fail a relaxed 5% gate.

Allen–Cahn MLP has median 2.330% and maximum 2.331%; both round to 2.33%.

Burgers–MLP has no seed filtering ($8/8$ admitted); the other nonlinear families are selected samples. A common 2% threshold retains $13/17$ runs; Appendix A reports the saved-output comparison on that subset.

Table 1 includes every attempted setting. A small training residual does not ensure an accurate field: one Kuramoto–Sivashinsky run has loss 4.4×10^{-5} at 103.8% non-feature error. The rejected settings therefore measure failure of this training-and-collocation protocol, consistent with known stiff-problem difficulties (Krishnapriyan et al., 2021; Wang et al., 2022).

Working likelihood, prior and centres. Write $r(\theta) \in \mathbb{R}^N$ for the stacked training residuals $r_{\mathrm{blk}} = w_{\mathrm{blk}} e_{\mathrm{blk}}$. The fixed factors are $w = 10$ on initial conditions, $w = 10$ on Burgers Dirichlet boundaries or Allen–Cahn periodic value matching, and $w = 1$ on collocation and periodic derivative matching. With common residual variance v, the working likelihood is $r(\vartheta) \sim \mathcal{N}(0, vI)$; hence the baseline negative log-likelihood is

$$-\log p = \tfrac{1}{2}\|r(\vartheta)\|^2 + \text{const} = \tfrac{1}{2}\sum_{\mathrm{blk}} w_{\mathrm{blk}}^2 \|e_{\mathrm{blk}}(\vartheta)\|^2 + \text{const} \qquad (v = 1).$$

The weights specify the likelihood as well as the training loss: the unweighted block noise standard deviations are $\sqrt{v}/w_{\mathrm{blk}}$. All analyses retain this raw weighted metric, with $J = \partial r/\partial\theta$ and $F = J^\top J$. For $\alpha, v > 0$, the isotropic prior is $\vartheta \sim \mathcal{N}(0, \alpha^{-1} I)$. The baseline uses $v = 1$ and $\alpha_0 = 1/\langle\theta^2\rangle$ over the parameters carrying variance; this plug-in rule is not evidence maximisation.

Freeze the affine residual $r(\theta) + J(\vartheta - \theta)$ and affine output $u_\theta(x) + g(x)^\top(\vartheta - \theta)$, where $g = \nabla_\theta u_\theta$. Conditioning on zero residual gives

$$\Sigma(\alpha, v) = v(F + \gamma I)^{-1}, \qquad \sigma^2(x) = g(x)^\top \Sigma g(x), \qquad \gamma = \alpha v,$$

and the exact affine posterior mean

$$\delta_\gamma = -(F + \gamma I)^{-1}[J^\top r(\theta) + \gamma\theta], \qquad m_\gamma(x) = u_\theta(x) + g(x)^\top \delta_\gamma. \tag{1}$$

The prior is absent during training, so u_θ need not equal m_γ even at a stationary training solution. A θ-centred (checkpoint-centred) prior $\vartheta \sim \mathcal{N}(\theta, \alpha^{-1}I)$ instead gives

$$\delta_{L,\gamma} = -(F + \gamma I)^{-1} J^\top r(\theta), \qquad m_\gamma^\theta = u_\theta + g^\top \delta_{L,\gamma}.$$

Its covariance is unchanged. This prior is conditional on the fitted checkpoint. We compare its affine mean at the stated scales and available trained-centre roots, and evaluate $\widetilde{u}_\gamma^\theta = u_{\theta+\delta_{L,\gamma}}$ as a nonlinear forward diagnostic. The covariance remains at θ; this diagnostic is not a nonlinear posterior predictive mean. Prediction bounds are in Appendix C. At fixed γ, centres and uncertainty contrasts are unchanged by matched (α, v) choices, while widths scale as $\sqrt{v}$. No output-noise term is added when comparing latent uncertainty with the deterministic reference.

Baseline variances use the Woodbury identity

$$\sigma^2(x) = \alpha^{-1}\left[\|g\|^2 - (Jg)^\top (JJ^\top + \alpha I)^{-1} Jg\right].$$

None of the 94 baseline solves required added diagonal regularisation. Weak-precision calculations instead factor J directly by QR/SVD to avoid cancellation (Appendix C).

Regional diagnostics. For each predictive centre, write e for prediction minus u^* and $\langle\cdot\rangle_r$ for a regional mean.

Definition 3 (Contrast, target and gap)

$$R_\sigma^{\mathrm{rms}} = \sqrt{\frac{\langle\sigma^2\rangle_{\mathcal{S}}}{\langle\sigma^2\rangle_{\mathcal{B}}}}, \qquad R_{\mathrm{tar}} = \sqrt{\frac{\langle e^2\rangle_{\mathcal{S}}}{\langle e^2\rangle_{\mathcal{B}}}}, \qquad \rho = R_{\mathrm{tar}}/R_\sigma^{\mathrm{rms}}.$$

The secondary mean-width contrast is $\widehat{R}_\sigma = \langle\sigma\rangle_{\mathcal{S}}/\langle\sigma\rangle_{\mathcal{B}}$.

Ratios assume positive denominators. At fixed scales, all centres share the contrast; the target, gap and coverage differ. The identity $\rho = 1$ matches regional RMS contrasts, not pointwise calibration or absolute width. We also report

$$\overline{Z}_r = \langle|e|/\sigma\rangle_r, \qquad \mathrm{cov@}k_r = \langle\mathbf{1}\{|e|/\sigma \le z_k\}\rangle_r,$$

with $z_{68} = 1$, $z_{95} = 1.96$. Under a correctly specified centred Gaussian error model their expectations are $\sqrt{2/\pi} \approx 0.8$ and approximately $0.68, 0.95$. Throughout, sampled curves describe evaluated settings; refined roots locate the observed trained-centre crossings within retained numerical brackets. Direct root fields are compared with the coverage envelopes. A *best tested evidence setting* maximises a finite search, and a *consistency check* compares numerical representations. All spectral conclusions use the returned affine model; the checks do not certify weak-mode forward accuracy.

3 Results and Analysis

The three contributions concern regional allocation, last-layer expressivity and spectral weighting. We first compare uncertainty with realised error, then examine the features and curvature that determine these quantities.

configuration	admitted	R_σ^{rms}	R_{tar}	ρ
Burgers, MLP	$8/8$	6.37 ± 0.38	130.6 ± 82.4	20.5 ± 13.1
Burgers, Fourier	$3/8$	$[3.857, 3.959]$	$[15.77, 116.5]$	$[4.09, 29.55]$
Allen–Cahn, MLP	$3/5$	$[6.871, 7.731]$	$[21.58, 33.58]$	$[2.958, 4.888]$
Allen–Cahn, Fourier	$3/4$	$[5.818, 5.93]$	$[12.64, 17.37]$	$[2.165, 2.986]$
configuration	$\overline{Z}_{\mathcal{S}}$	$\overline{Z}_{\mathcal{B}}$	$\mathrm{cov@68}_{\mathcal{S}}$	$\mathrm{cov@68}_{\mathcal{B}}$
Burgers, MLP	0.319 ± 0.275	0.012 ± 0.003	0.923 ± 0.083	1.000 ± 0.000
Burgers, Fourier	$[0.3996, 0.724]$	$[0.02456, 0.1248]$	$[0.8227, 0.8912]$	$[0.9997, 1]$
Allen–Cahn, MLP	$[0.1845, 0.3296]$	$[0.02925, 0.03078]$	$[0.9708, 1]$	1
Allen–Cahn, Fourier	$[0.2533, 0.2893]$	$[0.02237, 0.0284]$	$[0.9958, 1]$	1

Table 2: Unit-noise, plug-in-precision baseline with trained centres. Burgers–MLP entries are mean $\pm$ population standard deviation over eight admitted seeds; other entries are observed ranges over three admitted runs. Counts are admitted/attempted. Ranges are descriptive, not confidence intervals. Calibration benchmarks are $\overline{Z} \approx 0.8$ and one-standard-deviation coverage ≈ 0.68.

window	R_σ^{rms}	ρ	$\overline{Z}_{\mathcal{S}}$	cov@68	cov@95
$t < 1/\pi$	2.65 ± 0.23	16.9 ± 14.5	0.10 ± 0.07	0.99 ± 0.01	1.00 ± 0.00
$t \geq 1/\pi$	10.75 ± 0.71	20.6 ± 13.1	0.81 ± 0.81	0.76 ± 0.27	0.87 ± 0.17

Table 3: Burgers–MLP baseline split at $t = 1/\pi$, with trained centres; mean $\pm$ population standard deviation over eight seeds. Coverage and $\overline{Z}$ concern the sharp region. Pre-formation points constitute 69% of the global sharp mask.

3.1 Regional Allocation, Scale Sensitivity, and Predictive Centres

The comparisons retain the training-weighted homoscedastic likelihood, original reference regions and trained-network admission decisions. Burgers–MLP supplies eight unfiltered seeds; the other nonlinear families are smaller selected samples.

Regional allocation and scale sensitivity. Every admitted nonlinear run has a baseline allocation deficit (Table 2).

For the reference run, $(R_\sigma^{\mathrm{rms}}, R_{\mathrm{tar}}, \rho) = (6.47, 118.58, 18.3)$. Across Burgers–MLP seeds, R_σ^{rms} varies by 6% (coefficient of variation), over $[6.00, 7.25]$, whereas $R_{\mathrm{tar}} \in [10.0, 238.2]$. All eight attempted seeds pass admission, so this family's deficit is not an admission-selection effect. Small bulk-error denominators amplify target variation; the bulk is also strongly over-dispersed at this baseline.

The global mask mixes pre- and post-formation behaviour (Table 3). The mean gap exceeds one in both windows, so early points are not its sole source. Post-formation sharp coverage varies over $[0.37, 1.00]$ and $\overline{Z}_{\mathcal{S}}$ over $[0.02, 2.43]$, while $R_\sigma^{\mathrm{rms}} \in [9.93, 12.37]$: similar contrasts can accompany very different calibration. Allen–Cahn has no sharp points before $t \approx 0.4$ under the same global-percentile rule.

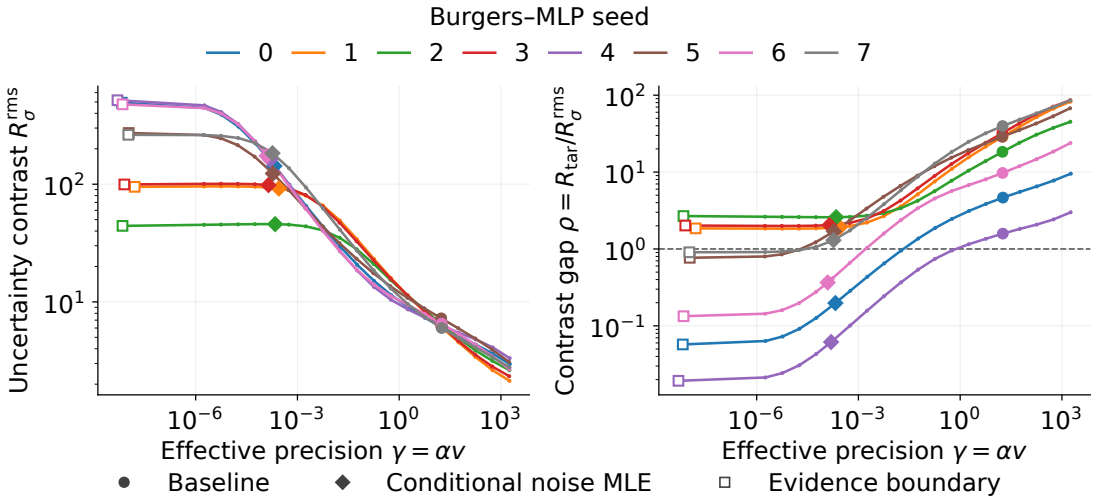


Figure 2: Uncertainty contrast (left) and gap $\rho = R_{\mathrm{tar}}/R_\sigma^{\mathrm{rms}}$ (right) for eight Burgers–MLP seeds with trained predictions fixed. Curves join 19 unit-noise prior settings and exact matched MLE/original-evidence points. Circles mark the baseline $(\alpha_0, 1)$, diamonds the conditional noise MLE $(\alpha_0, \widehat{v})$, and open squares the best original-grid evidence setting at the lower noise boundary. Selected settings share contrasts with their unit-noise aliases, but not absolute widths. The dashed line marks equal RMS contrasts, not calibration. Appendix C summarises other families, affine centres and extended-grid diagnostics; Appendix D gives their full curves.

The baseline contrast also changes by at most 1.1% between the two finest tested grids (Appendix A); Gaussian and Student-t distributions with matched covariance have identical RMS and mean-width contrasts.

The scale sweep covers all 17 admitted nonlinear checkpoints with residual weights and reference regions fixed, using trained and zero-mean-prior affine centres. Burgers also uses the global masks intersected with $t \geq 1/\pi$. The unit-noise sweep contains 19 prior multipliers from 10^{-7} to 10^2 and matched points for the conditional noise MLE and finite-grid evidence selections. A centre change preserves R_σ^{rms} at fixed γ, but changes the target and calibration diagnostics.

Five trained-centre Burgers–MLP seeds cross $\rho = 1$ in the observed brackets (Figure 2). Saved modal sums refine the crossings to a range from 2.05×10^{-5} (seed 5) to 0.736 (seed 4), more than four orders of magnitude apart in effective precision (Table 12). The original sampled brackets are retained in Appendix C. Seeds 1–3 remain above one, with minimum sampled gaps 1.84, 2.59 and 1.99 (Appendix C). Seed dependence lies in both quantities: baseline uncertainty contrast varies by 6%, while each seed's largest sampled contrast ranges from 45.83 to 517.34. The QR-complement ratios span $[42.71, 509.9]$ (Table 7), so weak-precision geometry also varies; they identify the whole-kernel limit only under the rank condition in Section 3.3.

Training-based scale choices. The conditional MLE $\widehat{v} = \|r(\theta)\|^2/N$ fixes the trained parameter. Fitted-residual reuse can underestimate variance; this plug-in residual scale is not an unbiased discrepancy estimate. No classical degrees-of-freedom correction is assigned here (Appendix E).

With trained centres it reverses the allocation deficit in seeds 0, 4 and 6; the other five have $\rho \in [1.30, 2.59]$. Nevertheless, all eight seeds under-cover both regions: one-standard-deviation coverage is 0.198–0.416 in bulk and 0.018–0.5315 in sharp; at 1.96 standard deviations it is 0.412–0.721 and 0.051–0.698.

On both zero-mean-prior evidence grids, $14/17$ best tested pairs lie at the lowest allowed noise variance; extending the range does not resolve those boundary selections. The three Allen–Cahn Fourier candidates are interior, whereas singular-value perturbations substantially change Allen–Cahn MLP selections. These are finite-grid candidates rather than reliably identified scales. Appendix C records the projected-data mechanism, extension and numerical sensitivity.

Predictive centres and linearisation. At the baseline, the zero-mean-prior affine centre increases full-grid error RMS in $14/17$ checkpoints (Table 9). For Burgers–MLP, mean R_{tar} falls from 130.6 to 13.13 and mean ρ from 20.47 to 2.09, but mean sharp error RMS rises from 0.1138 to 0.2448 and bulk error RMS from 0.000805 to 0.01821; sharp error worsens in seven of eight seeds. Thus a reduced target can accompany worse prediction.

The displacement separates as $\delta_\gamma = \delta_{L,\gamma} + \delta_{P,\gamma}$, where the likelihood term is the shift under the θ-centred prior and $\delta_{P,\gamma} = -\gamma(F + \gamma I)^{-1}\theta$ is prior shrinkage. Its QR-complement term is $-\theta_c$ at every scale, where θ_c is the parameter projection onto that complement. At the baseline, total displacement is 99.78–99.96% of $\|\theta\|$, whereas the likelihood term is at most 0.0155%. Prediction-level bounds establish prior-only worsening in all 14 affected runs (Appendix C); parameter norms alone would not establish that attribution.

For the θ-centred prediction, the pointwise baseline shift bound ranges from 0.00749 to 0.0908 standard deviations. Bounds establish $\rho > 1$ in $16/17$; direct evaluation gives $\rho > 1$ in all 17, resolving seed 4 with gap 1.582. Both regions over-cover at one standard deviation. At the same conditional-MLE scales, $\rho^\theta \in [2.18, 10.01]$. The affine mean improves full-grid RMS error in $15/17$ runs (Table 9), with median error ratio 0.7953. At one standard deviation, all 17 bulk regions over-cover and $12/17$ sharp regions under-cover. Thus prediction can improve without regional calibration. The residual variance is not re-estimated after shifting.

At the baseline, nonlinear evaluation $u_{\theta+\delta_{L,\gamma}}$ also retains $\rho > 1$ in all 17, with regional RMS affine remainder below 6.4×10^{-4} of RMS σ. At the MLE, however, the bulk remainder is 2.495–386.16 times RMS σ, despite $\|\delta_L\|/\|\theta\| \in [0.00782, 0.03867]$. Full-grid error worsens in $11/17$ and bulk error in all 17. The nonlinear gaps fall below one through bulk deterioration, not repaired allocation. These are forward diagnostics with covariance frozen at θ.

Oracle: contrast agreement and coverage. A descriptive oracle rescales each saved covariance to target 68% bulk coverage using the higher empirical quantile of $|e|/\sigma$, separately for each centre and time subset. It uses reference errors and is not a practical tuning rule. For the five trained-centre bracketing seeds, the closest saved points have $\rho \in [0.954, 1.032]$, yet sharp coverage is 0.343–0.652 at one standard deviation and 0.692–0.864 at 1.96. For zero-mean-prior affine post-formation predictions, the closest points across all eight seeds have $\rho \in [0.933, 1.193]$ and sharp one-standard-deviation coverage 0.009–0.347. At the five refined trained-centre roots, direct sharp coverage is 0.30325–0.55725 at one standard deviation and 0.67725–0.852 at 1.96 (Table 12). All values lie within the positive-modal envelopes, whose upper bounds are at most 0.563 and 0.862, respectively (Appendix C). Contrast agreement therefore does not restore sharp coverage under this bulk calibration.

At these five trained-centre roots, the θ-centred affine prediction has full-grid error RMS 0.493–0.973 times that of the trained prediction and $\rho^\theta \in [1.13, 7.58]$; these precisions do not match its

own error contrast. After its own bulk-68% calibration, sharp one-standard-deviation coverage is 0.1165–0.424.

For $Z_j = e_j/\sigma_j$, define the variance-weighted regional second moment

$$M_r = \frac{\sum_{j\in r}\sigma_j^2 Z_j^2}{\sum_{j\in r}\sigma_j^2} = \frac{\langle e^2\rangle_r}{\langle\sigma^2\rangle_r}, \qquad \rho^2 = M_{\mathcal{S}}/M_{\mathcal{B}}.$$

Thus $\rho = 1$ equates weighted second moments, leaving ordinary coverage quantiles undetermined. At the five nearest saved oracle settings, using bulk-calibrated widths, sharp median $|Z|$ ranges over $[0.725, 1.273]$, compared with $[0.655, 0.701]$ in bulk; sharp 99th percentiles are 3.60–8.70, versus 3.05–4.66 in bulk. Sharp signed means include -1.96 (seed 5) and 0.85 (seed 7). The top 1% of sharp points carry 15.3–23.8% of error energy, and width–absolute-error rank associations are 0.693–0.936. Larger sharp medians show that the difference extends beyond extreme tails; regional sign bias and substantial width–error ordering coexist with undercoverage.

An RMS deficit can coexist with overcoverage: seed 1 has closest gap 1.836 and sharp one-standard-deviation coverage 0.8605; seed 3 has gap 1.992 and coverage 0.725 (Tables 11 and 13).

Sharp calibration using its higher empirical 0.68 quantile gives sharp 1.96-standard-deviation coverage 0.8345–0.95775 for Burgers–MLP, below 0.95 in $7/8$ runs. Other families and time subsets include overcoverage (Table 13): matching one quantile need not match another. This output rescaling preserves within-region width patterns and does not test a residual-informed likelihood that changes $J^\top D^{-1} J$.

In the sampled band $0.9 \le \rho \le 1.1$, Burgers–MLP seed 7 has seven settings with sharp one-standard-deviation coverage 0.44725–0.986 after bulk calibration: $\rho = 0.925$ gives 0.756, while $\rho = 1.064$ gives 0.44725. These include undercoverage and overcoverage, without a continuous-band claim (Appendix C).

Ensemble comparison. An ensemble (Lakshminarayanan et al., 2017) of the eight Burgers–MLP runs, with spread computed at divisor $K = 8$, has $R_\sigma^{\mathrm{rms}} = 136.5$, its own mean-prediction target $R_{\mathrm{tar}} = 158.2$ and $\rho = 1.16$. Sharp $\overline{Z} = 0.64$ and coverage is $0.876/0.993$ at $68/95\%$, while bulk $\overline{Z} = 0.45$. It nearly matches allocation but does not restore absolute calibration. The ordering against individual Gaussians depends on their scale: at the conditional MLE three trained-centre gaps are below one and five lie in $[1.30, 2.59]$, each relative to its own predictor's error. Omitting one ensemble member gives gaps 1.055–1.351, sharp 68% coverage 0.4935–0.9075 and 95% coverage 0.9720–0.9995. Omitting seed 5 gives the largest gap and lowest 68% coverage; its sharp error has correlation -0.7517 with the other seven's mean error. These ranges measure membership sensitivity, not confidence intervals: each omission changes the predictor, target and spread.

3.2 Feature Expressivity for the Reference Solution

The trained network's 129 last-layer features, including the bias, can approximate the sharp-region reference accurately when their readout is refitted using reference values. This tests feature expressivity, separately from the coefficients selected by residual-only training or the affine posterior. Let Φ be the feature matrix on the $80{,}000$ evaluation points and $e = u_\theta - u^*$ the sampled trained-network error.

Definition 4 (Per-region fraction and masked-global ratio) *For a region r with nonzero error e_r and local feature matrix Φ_r, let P_r denote the orthogonal projector onto* $\mathrm{col}(\Phi_r)$ *and define the* per-region *fraction*

$$f_r^{\perp} \;=\; \frac{\|e_r - P_r e_r\|^2}{\|e_r\|^2}. \tag{2}$$

Let P instead denote the projector onto $\mathrm{col}(\Phi)$ *formed once globally, and define the* masked-global *ratio*

$$\tilde{f}_r^{\perp} \;=\; \frac{\big\|(e - Pe)\big|_r\big\|^2}{\|e_r\|^2}.$$

The linear readout gives $u_\theta|_r \in \mathrm{col}(\Phi_r)$. Since $e_r = u_\theta|_r - u^*|_r$,

$$(I - P_r)e_r = -(I - P_r)u^*|_r, \qquad f_r^{\perp} = \frac{\|(I - P_r)u^*|_r\|^2}{\|e_r\|^2}.$$

The numerator is local reference-approximation error; the denominator is trained-network error. Their regional ordering therefore depends on both.

Remark 5 (The per-region diagnostic is a fraction) *For any self-adjoint idempotent map P and nonzero e, $\|e\|^2 = \|Pe\|^2 + \|e - Pe\|^2$, so $f_r^{\perp} \in [0, 1]$.*

Proof Self-adjointness and idempotence give $\langle e, Pe\rangle = \|Pe\|^2$; expanding $\|e - Pe\|^2$ gives the identity and both bounds. ■

Remark 6 (The masked-global quantity is not a fraction) *The masked-global ratio can exceed one, because global orthogonality need not survive restriction to a region.*

Proof For $\varphi = (1, 1)^\top$, $e = (1, 1/10)^\top$ and $r = \{2\}$, the rank-one projection is $P_\varphi e = (11/20)\varphi$, giving $\tilde{f}_r^{\perp} = 81/4 = 20.25$. ■

For comparison, $\tilde{f}_{\mathcal{B}}^{\perp} = 3.30$ on Burgers–MLP seed 2 and 4.4 on a synthetic control with error concentrated in a 5% subset.

Measured off-span fractions. Across all 17 runs, $f_{\mathcal{S}}^{\perp}$ is the smallest regional fraction. Its maximum is 0.011616 (Burgers–MLP seed 4): at least 98.8% of sampled sharp error lies in the local last-layer span. The bulk-to-sharp fraction ratio ranges from 9.37 to 542.5 (the displayed runs span 18–266). Span containment makes this residual an upper bound for the full-network projection residual on the same points.

Held-out fits. Within each region we fit alternating ten-slice time blocks and test the remaining blocks, then swap, using unregularised least squares of numerical rank 129. The pooled test residual energy, divided by total regional error energy, is below 4% in $16/17$ sharp-region fits (median 0.72%). The worst fraction, 14.2%, is on Burgers–MLP seed 4, whose sharp error RMS 0.00362 is the smallest among Burgers–MLP seeds; its test residual RMS is 0.00136, versus error RMS 0.283 on seed 2. Random halves remain close to same-grid fits (Table 4), but blocked transition residual exceeds error energy in $10/17$ runs. Because the trained prediction u_θ lies in the last-layer span and

	Same grid: $f_r^{\perp}$			Held-out sharp region	
configuration	$\mathcal{B}$	$\mathcal{T}$	$\mathcal{S}$	random	10-slice blocks
Burgers, MLP (seed 2)	0.002	0.001	**0.0001**	0.000146	0.000455
Burgers, Fourier FS $= 2$ (seed 2)	0.071	0.057	**0.0015**	0.00163	0.00670
Allen–Cahn, MLP (seed 0)	0.230	0.030	**0.0009**	0.00139	0.0324
Allen–Cahn, Fourier FS $= 1$ (seed 0)	0.051	0.017	**0.0007**	0.000841	0.00987

Table 4: Reference expressivity of 129 last-layer features, including the bias. Residual energies are normalised by trained-network regional error. Same-grid columns give (2); held-out columns pool both test directions for random halves and alternating ten-slice time blocks.

each fit has full column rank, fitting e or u^* yields the same held-out residual energy. The pooled held-out residuals thus measure reference approximation by the local feature span, using additional reference information. These are local last-layer tests within the evaluation grid; span containment transfers only the same-grid bound, not held-out performance or a single global coefficient vector.

Expressivity and coefficient selection. Regional refits use reference values, whereas (1) selects coefficients from the residual likelihood and zero-mean parameter prior. The predictive-centre analysis in Section 3.1 attributes its prediction worsening to prior shrinkage. Together, these diagnostics distinguish feature expressivity from coefficient selection; they do not identify the cause of the trained error or its uncertainty mismatch.

3.3 Spectral Weighting of Regional Variance

Covariance allocation is a weighted sum of tangent energies. Its modal quotient connects the scale curves to the trained features; coarse spectral bounds describe what remains identifiable after grouping those modes.

Modal weights and limits. Let $F = V\Lambda V^\top$, with eigenvalues $\lambda_i \geq 0$, largest eigenvalue $\lambda_{\max}$ and orthonormal eigenvectors v_i, and write $c_i(x) = g(x)^\top v_i$. At unit noise,

$$\sigma^2(g) = g^\top(\alpha I + F)^{-1} g = \sum_i \frac{c_i^2}{\lambda_i + \alpha}. \tag{3}$$

For residual variance v, each term is $vc_i^2/(\lambda_i + \gamma)$, $\gamma = \alpha v$. Defining $a_{r,i} = \langle c_i(x)^2 \rangle_r$ gives

$$(R_\sigma^{\mathrm{rms}}(\gamma))^2 = \frac{\sum_i a_{\mathcal{S},i}/(\lambda_i + \gamma)}{\sum_i a_{\mathcal{B},i}/(\lambda_i + \gamma)}. \tag{3a}$$

For these finite sums,

$$\lim_{\gamma\to\infty} R_\sigma^{\mathrm{rms}} = \sqrt{A_{\mathcal{S}}/A_{\mathcal{B}}}, \quad A_r = \langle \|g\|^2 \rangle_r, \qquad \lim_{\gamma\downarrow 0} R_\sigma^{\mathrm{rms}} = \sqrt{K_{\mathcal{S}}/K_{\mathcal{B}}}, \quad K_r = \langle \|\Pi_{\ker J} g\|^2 \rangle_r,$$

provided $A_{\mathcal{B}} > 0$ and $K_{\mathcal{B}} > 0$, respectively. Multiplication by γ gives both termwise limits. The QR-complement ratio equals the whole-kernel ratio when J has full row rank. For $A_{\mathcal{S}}, A_{\mathcal{B}} > 0$, at strong precision, writing $\mu_r = \sum_i a_{r,i}\lambda_i / A_r$ gives the more precise expansion

$$(R_\sigma^{\mathrm{rms}}(\gamma))^2 = \frac{A_{\mathcal{S}}}{A_{\mathcal{B}}} \left[1 - \frac{\mu_{\mathcal{S}} - \mu_{\mathcal{B}}}{\gamma} + O\left(\frac{\lambda_{\max}^2}{\gamma^2} \right) \right]. \tag{4}$$

This follows by expanding each $(\lambda_i+\gamma)^{-1}$ and dividing the two sums. Neither endpoint bounds intermediate contrast, and monotonicity need not hold. For example, $\lambda=(0,1,100)$, $a_{\mathcal{B}}=(1,1,100)$ and $a_{\mathcal{S}}=(1,100,1)$ give both limits equal to 1, but $R_\sigma^{\mathrm{rms}}(1)=\sqrt{10304/503}\approx 4.53$. The measured curves and strong-precision checks are in Appendix C. The conditional limits and counterexample are Lean-verified (Table 18); expansion (4) remains outside verification. Figure 2 changes the modal weights while holding the trained tangents fixed. The predictive centres share the covariance contrast but have different error targets.

Two-subspace bounds. We group the modes by a spectral cutoff.

Definition 7 (Spectral split and regional loadings) *At threshold $\epsilon>0$, let Π_{sl} project onto modes with $\lambda_i\le\epsilon$ and Π_{st} onto those with $\lambda_i>\epsilon$. Define*

$$Q_{r,\mathrm{sl}}=\langle\|\Pi_{\mathrm{sl}}g\|^2\rangle_r,\qquad Q_{r,\mathrm{st}}=\langle\|\Pi_{\mathrm{st}}g\|^2\rangle_r.$$

For an individual vector, the lower and upper envelopes are

$$\begin{aligned} m_\epsilon(g)&=\frac{\|\Pi_{\mathrm{sl}}g\|^2}{\alpha+\epsilon}+\frac{\|\Pi_{\mathrm{st}}g\|^2}{\alpha+\lambda_{\max}},\\ M_\epsilon(g)&=\frac{\|\Pi_{\mathrm{sl}}g\|^2}{\alpha}+\frac{\|\Pi_{\mathrm{st}}g\|^2}{\alpha+\epsilon}. \end{aligned}\tag{5}$$

Theorem 8, proved in Appendix B, bounds each pointwise variance between these envelopes. Averaging its termwise inequalities gives the primary regional bound

$$L_{\mathrm{reg}}(\epsilon)\le(R_\sigma^{\mathrm{rms}})^2\le U_{\mathrm{reg}}(\epsilon).\tag{6}$$

Here $L_{\mathrm{reg}}(\epsilon)=\langle m_\epsilon(g)\rangle_{\mathcal{S}}/\langle M_\epsilon(g)\rangle_{\mathcal{B}}$ and $U_{\mathrm{reg}}(\epsilon)=\langle M_\epsilon(g)\rangle_{\mathcal{S}}/\langle m_\epsilon(g)\rangle_{\mathcal{B}}$, assuming positive total bulk energy. This is exact finite-sum aggregation, without a Jensen step. For common variance v, replace α by γ; the common factor cancels. The envelopes are sharp over spectra compatible with these summaries (Proposition 9, proved in Appendix B), but they discard energy placement within each block. They can therefore remain wide even after intersecting thresholds. Appendix B treats secondary aggregations; Appendix F, Table 18, records the formal verification scope.

Measured spectrum and loadings. The canonical Gram eigensolves return positive eigenvalues spanning 19.6–21.2 decades, but values below $\varepsilon_{64}\lambda_{\max}$ are not individually resolved. Their inverse lifts $J^\top w_i/\sqrt{\lambda_i}$ are unstable near that scale. We instead resolve the separated strong subspace and compute

$$\|\Pi_{\mathrm{sl}}g\|^2=\|g\|^2-\|\Pi_{\mathrm{st}}g\|^2.$$

This includes Fisher-null directions without division by small eigenvalues. Orthonormality, cutoff separation, eigenpair error and independent variance checks are recorded in Appendix A.

Sharp-to-bulk tangent-energy ratios are close to one on Burgers: bulk/transition/sharp means are $134/136/141$ for MLP and $114/119/120$ for Fourier. The canonical strong- precision RMS limits are $1.02, 1.03, 1.46, 1.09$, whereas baseline configuration-mean contrasts are 3.92–7.30; covariance weighting supplies the amplification. At $\epsilon=\alpha$, the sharp field loads the weak subspace 43.7–259.5 times more strongly than bulk (Table 5). All 11 evaluated baseline brackets contain their measured RMS contrasts but are too wide to establish a deficit relative to the error target. For eight Burgers–MLP seeds, primary loading ratios span 171.2–268.0; supporting mean-tangent ratios span 380.6–742.0.

configuration	returned decades	below roundoff	$Q_{\mathcal{S},\mathrm{sl}}/Q_{\mathcal{B},\mathrm{sl}}$	$M_{\mathcal{S},\mathrm{sl}}/M_{\mathcal{B},\mathrm{sl}}$	RMS bracket
Burgers–MLP, seed 2	21.2	82%	259.5	514.9	$[0.0488, 307.86]$
Burgers–Fourier, seed 2	19.9	13%	43.7	195.4	$[0.0648, 103.51]$
Allen–Cahn–MLP, seed 0	20.4	66%	59.3	268.6	$[0.0982, 113.87]$
Allen–Cahn–Fourier, seed 0	19.6	26%	79.0	1132.4	$[0.1743, 55.92]$

Table 5: Canonical spectrum and baseline sub-α loadings. Returned decades include unresolved positive eigenvalues; the next column is the fraction below $\varepsilon_{64}\lambda_{\max}$. Primary loading uses $Q_{r,\mathrm{sl}} = \langle \|\Pi_{\mathrm{sl}} g\|^2 \rangle_r$; the supporting mean-tangent loading uses $M_{r,\mathrm{sl}} = \|\Pi_{\mathrm{sl}} \langle g \rangle_r\|^2$. The final column gives (6) at $\epsilon = \alpha$, in contrast units, rounded outward from recorded energies. Only Burgers–MLP has replicated baseline loadings.

The full-grid study extends this geometry to all 17 runs. Among the full-time sampled curves, $13/17$ have both increasing and decreasing segments, and $12/17$ leave the interval formed by the tangent and QR-complement ratios. The saved fields reach $\gamma/\lambda_{\max} = 6.28 \times 10^{-5}$–$2.96 \times 10^{-4}$. Evaluation of their regional modal sums at $10^3\lambda_{\max}$ confirms approach to the tangent-norm limit, within 1.94×10^{-4} relative across all full and Burgers post-formation curves. Small positive singular values are retained without exact-null classification. The measured curves, conditional limits and coarse brackets answer distinct questions (Appendix C).

4 Discussion and Conclusion

Regional allocation and coverage are distinct properties of this post-hoc Gaussian. Likelihood and scales determine covariance allocation; the predictive centre also changes the error target and coverage. The tested training-only choices do not consistently calibrate both regions.

Predictive centre and linearisation. The baseline mismatch survives removal of zero-mean prior shrinkage. At weaker precision, affine prediction can improve without regional calibration, and a modest relative parameter shift can produce a large nonlinear remainder.

Held-out refits demonstrate feature expressivity, separately from coefficient selection. Spectral weights explain scale dependence, but the conditional limits do not bound intermediate contrast, and coarse envelopes cannot certify the baseline deficit. Evidence boundaries and perturbation sensitivity leave weak-precision scale selection unresolved.

Residual-informed likelihoods. Residual-informed or block-wise variances D replace F by $J^\top D^{-1} J$, after factoring out global noise variance, changing spectral directions and within-region allocation. The common-scale oracle does not test this model. The rejected Kuramoto–Sivashinsky example links small aggregate residuals to large solution errors, without ruling out useful local residual information. Such likelihood design remains untested.

Limitations. The study covers two $1{+}1$-dimensional nonlinear equations and two architectures with a shared sub-network/fusion-head design. Only Burgers–MLP supplies eight fully admitted seeds; the remaining families are small selected samples. Admission thresholds, raw residual weights and reference-based regions are experimental conventions. Root-coverage envelopes concern the five observed trained-centre crossings and the stated bulk calibration, rather than optimal joint calibration.

The θ-centred prior is conditional on fitted parameters. Nonlinear forward checks retain covariance at the original checkpoint; they do not establish a nonlinear posterior approximation. Spectral calculations do not certify weak-mode forward accuracy or exact rank. The synthetic stress test exceeds its whole-grid evidence tolerance, despite agreement at selected settings (Appendix C). Held-out expressivity is last-layer and within-grid; it does not establish off-grid or full-network held-out approximation. Transfer to higher dimensions, other architectures and training objectives is untested.

Implications. Assess accuracy, regional contrast and coverage for each predictive centre; ensemble sensitivity likewise warrants assessment of the particular ensemble. The trained-centre Gaussian is a covariance diagnostic. An archived random half-grid split-conformal illustration (Lei et al., 2018) on the reference run, using absolute-error scores at target 0.95, gives marginal coverage 0.950 but bulk, transition and sharp coverage 0.98, 0.99 and 0.31. This descriptive finite-grid example shows how marginal coverage can conceal regional imbalance.

Code and data availability. Code, trained parameters, saved results and the Lean development are available at `https://github.com/soumya1729/pinn-uq-scale-sensitivity`. The tagged reproduction release supplies the saved-data archive and its checksum.

Conclusion. Under the specified likelihood, sharp intervals under-cover at all five trained-centre roots despite bulk calibration and regional RMS contrast agreement. Allocation and coverage therefore require separate regional checks for each run and predictive centre.

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Appendix A. Experimental and Reporting Details

The following details supplement the model and admission rule in Section 2. Training uses float64 throughout. Each architecture has 20 spatial and 20 temporal sub-networks of widths $[\,\cdot\,, 64, 64, 1]$ feeding a $[40, 128, 128, 1]$ fusion head. Activations are $\tanh$, biases are zero and weights are drawn from $\mathcal{N}(0, 2/d_{\mathrm{in}})$. Fourier models use 16 random features per axis, with bandwidth 2 for Burgers and 1 for Allen–Cahn. Adam has learning rate 10^{-3}, with 5000 Burgers steps and $15{,}000$ Allen–Cahn steps, followed by 50 and 200 Levenberg–Marquardt steps. Each LM step uses at most 100 CG iterations, initial damping 10^{-4} and minimum 10^{-12}; it accepts at $\rho_{\mathrm{LM}} > 0.25$, decreases damping by 3 at $\rho_{\mathrm{LM}} > 0.75$ and multiplies it by 4 on rejection, allowing at most 5 consecutive rejections. Residual row counts and weights are given in Section 2; complete implementation settings are preserved with the code.

Reported multi-run means average per-run quantities. Every $\pm$ denotes the population standard deviation with divisor n; ranges give observed minima and maxima. The three-run control varies paired pulse-width/diffusivity settings at seed 0. Mean gaps are means of ratios, not ratios of mean contrasts: on Burgers–MLP they are 20.47 and 20.49, respectively. Reference-error oracles are evaluation devices, not training-based tuning rules.

A common 2% admission cutoff retains $13/17$ of the original cohort: $8/8$ Burgers–MLP, $3/3$ Burgers–Fourier, Allen–Cahn MLP seed 3 and Allen–Cahn Fourier seed 1. The original 3% Allen–Cahn cutoff remains a convention, not a PDE-derived tolerance. The unchanged masks and saved fields permit the subset comparison below; its two Allen–Cahn results are single-run descriptions. In this subset, the baseline zero-mean affine error increases in $10/13$ runs; direct θ-centred baseline predictions give $\rho^\theta > 1$ in all 13, including the one case unresolved by the $12/13$ bound. With trained centres, conditional-MLE sharp coverage is below nominal in all 13 runs: $[0.018, 0.62425]$ at one sigma and $[0.051, 0.768]$ at 1.96 sigma. Bulk coverage is below nominal in $10/13$ and $12/13$ runs at those thresholds, respectively. All eight Burgers–MLP findings and the five refined trained-centre crossings remain in this subset by construction.

Linear-control context. The contextual linear control solves $u_t + au_x = \nu u_{xx}$ with $a = 1$ and a periodic Gaussian pulse. Three of four MLP attempts pass admission; the admitted seed-0 runs vary paired pulse-width/diffusivity settings. At the baseline their uncertainty contrasts are $[1.086, 1.331]$, error contrasts $[0.7816, 0.9986]$ and gaps $[0.6521, 0.899]$. They are not part of the nonlinear 17-run scale study.

Grid and spectral checks. The top-5% region rule alone does not establish feature resolution. For the sharpest slice, the discrete integral width in cells is

$$W_{\text{cells}} = \frac{\sum_j |\partial_x u^*(x_j, t)|}{\max_j |\partial_x u^*(x_j, t)|}.$$

This separates a one-cell spike from a resolved layer, whereas half-maximum crossing counts have a discrete floor. The width exceeds one cell at the tested finest grids; primary RMS and secondary mean-width contrasts change by at most 1.1% between $n_x = 400$ and 200. The canonical Gram spectral projectors were checked for orthonormality, separation at the cutoff and eigenpair backward error; independent Woodbury evaluations checked variances. The all-17 direct-J checks are summarised in Appendix C, with numerical protocols in Appendix E.

Appendix B. Spectral Bounds and Aggregation

The following statements retain the proofs supporting the two-subspace summaries in Section 3.3 and their aggregation to regional statistics.

Theorem 8 (Two-sided spectral sandwich) *For $F \succeq 0$, $\alpha, \epsilon > 0$ and $g_{\mathrm{b}} \neq 0$,*

$$L(\epsilon) := \frac{m_\epsilon(g_{\mathrm{s}})}{M_\epsilon(g_{\mathrm{b}})} \leq \frac{\sigma^2(g_{\mathrm{s}})}{\sigma^2(g_{\mathrm{b}})} \leq \frac{M_\epsilon(g_{\mathrm{s}})}{m_\epsilon(g_{\mathrm{b}})} =: U(\epsilon). \tag{7}$$

Consequently, the ratio lies between $\sup_{\epsilon>0} L(\epsilon)$ and $\inf_{\epsilon>0} U(\epsilon)$.

Proof In the weak block, $\alpha \leq \alpha + \lambda_i \leq \alpha + \epsilon$; in the strong block, $\alpha + \epsilon < \alpha + \lambda_i \leq \alpha + \lambda_{\max}$. Applying reciprocal bounds to (3) gives $m_\epsilon(g) \leq \sigma^2(g) \leq M_\epsilon(g)$. Both bulk envelopes are positive, so division in opposite directions gives the result. ■

The primary statistic averages variance before taking square roots. The secondary mean-width ratio is $\widehat{R}_\sigma = \langle \sigma(g) \rangle_{\mathcal{S}} / \langle \sigma(g) \rangle_{\mathcal{B}}$. Uniform pairwise bounds also transfer to it: if $\ell \sigma(g_b) \leq \sigma(g_s) \leq$

$u\sigma(g_b)$ for every sharp–bulk pair, averaging over both finite regions and dividing by positive bulk mean width gives $\ell \le \widehat{R}_\sigma \le u$. This differs from evaluating variance at a region-mean tangent.

For regions with positive mean width, define $\kappa_r = \sigma(\langle g\rangle_r)/\langle\sigma(g)\rangle_r$. The covariance norm triangle inequality gives $0 \le \kappa_r \le 1$, and cancellation of the two mean widths gives, when $\kappa_{\mathcal{B}} > 0$,

$$\frac{\sigma(\langle g\rangle_{\mathcal{S}})}{\sigma(\langle g\rangle_{\mathcal{B}})} = \frac{\kappa_{\mathcal{S}}}{\kappa_{\mathcal{B}}}\widehat{R}_\sigma. \tag{8}$$

Equation (8) shows why mean-tangent loadings cannot replace the primary pointwise energies. The following sharpness result concerns the information in those energies.

Sharpness of the spectral bounds. The following construction shows what can be inferred from the four block energies alone. It does not assert that measured networks attain the bounding envelopes.

Proposition 9 (Sharpness) *Fix $\alpha > 0$, $0 < \epsilon < \lambda_{\max}$ and four nonnegative block energies, with positive total bulk energy. Spectra and vectors realising these energies can approach either endpoint of* (7)*.*

Proof For $\epsilon < \epsilon' \le \lambda_{\max}$, use four orthogonal modes with eigenvalues $0, \epsilon, \epsilon', \lambda_{\max}$. To approach U, place the sharp weak and strong energies at $0, \epsilon'$, and the bulk energies at $\epsilon, \lambda_{\max}$. Square-root coordinates realise the prescribed energies and give

$$\frac{q_{\mathrm{s,sl}}/\alpha + q_{\mathrm{s,st}}/(\alpha+\epsilon')}{q_{\mathrm{b,sl}}/(\alpha+\epsilon) + q_{\mathrm{b,st}}/(\alpha+\lambda_{\max})} \longrightarrow U(\epsilon) \quad (\epsilon' \downarrow \epsilon).$$

For L, place sharp energies at $\epsilon, \lambda_{\max}$ and bulk energies at $0, \epsilon'$. The ratio then tends to $L(\epsilon)$. The strict inequality $\epsilon' > \epsilon$ keeps that mode in the strong block. ∎

Appendix C. Full-Grid Scale and Centre Diagnostics

These diagnostics use all 17 admitted checkpoints: Burgers–MLP seeds 0–7, Burgers–Fourier seeds $0, 2, 6$, and Allen–Cahn MLP/Fourier seeds $0, 1, 3$. Trained parameters, raw weighted residuals, reference fields, 400×200 grids and global percentile masks remain fixed. Recentring does not change admission. Affine means use the frozen output; $u_{\theta+\delta_L}$ is a separate nonlinear diagnostic with covariance still at θ. The three-run families are exploratory. Appendix C retains per-seed Burgers–MLP results; Appendix D gives extended curves and tables.

Burgers–MLP per-seed results. Table 6 retains the eight trained-centre runs. Its minima use the original grid and matched aliases, whereas the all-17 diagnostics also include expanded evidence and sensitivity-selected settings.

Approximate sampled crossing brackets in αv are

$$\begin{aligned}
&\text{seed } 0: \ [0.0182150, 0.0576010], && \text{seed } 4: \ [0.569767, 1.80176],\\
&\text{seed } 5: \ [1.73345, 5.48164] \times 10^{-5}, && \text{seed } 6: \ [5.61912, 17.7692] \times 10^{-4},\\
&\text{seed } 7: \ [1.78416, 5.64201] \times 10^{-5}.
\end{aligned}$$

These intervals need not contain unique roots. No sampled crossing is found for seeds 1–3; this does not exclude unsampled crossings.

seed	baseline ρ	MLE ρ	min. sampled ρ	bulk coverage 68%	bulk coverage 95%	sharp coverage 68%	sharp coverage 95%
0	4.66	0.20	0.06	0.261	0.501	0.516	0.684
1	28.83	1.91	1.84	0.274	0.455	0.095	0.273
2	18.33	2.59	2.59	0.380	0.598	0.137	0.349
3	32.10	2.04	1.99	0.198	0.412	0.143	0.339
4	1.58	0.06	0.02	0.233	0.449	0.531	0.630
5	28.82	1.70	0.77	0.416	0.721	0.018	0.051
6	9.74	0.37	0.13	0.247	0.459	0.363	0.698
7	39.69	1.30	0.91	0.356	0.617	0.060	0.161

Table 6: Original Burgers–MLP scale study at the fixed trained centre. Baseline and MLE gaps use $(\alpha_0, 1)$ and $(\alpha_0, \widehat{v})$. Minima use the original 19 unit-noise prior multipliers and matched MLE/evidence points. Coverage is at the conditional MLE, within one and 1.96 standard deviations.

Settings and computation. The unit-noise sweep uses 19 prior multipliers 10^{-7}–10^{2}, with exact aliases for conditional-MLE and evidence-selected effective precisions. The evidence grids span $\alpha/\alpha_0 = 10^{-7}$–$10^{7}$ and initially $v/\widehat{v} = 10^{-4}$–10^{6} in quarter decades; one extension lowers the noise edge to $10^{-8}\widehat{v}$. Six prescribed spectral sensitivity cases supply further settings. For each prior centre, shared $\gamma = \alpha v$ gives the same contrast and affine mean; widths differ by $\sqrt{v}$, without added predictive noise.

One direct-J Householder QR/SVD per checkpoint gives $J^\top = Q_1 U \operatorname{diag}(s_i) V^\top$, with $Q = [Q_1, Q_c]$. All $N = 6400$ singular directions and $P - N$ complement coordinates are retained, without a Gram matrix, rank cutoff or noise floor. Prediction/loss reconstruction and baseline width comparisons give maximum width relative discrepancy 5.41×10^{-11}. At all 401 settings, independent differentiation checks $h_\gamma = J^\top(r + J\delta_\gamma) + \gamma(\theta + \delta_\gamma)$. The maximum normwise backward error is 6.19×10^{-17} against a 10^{-7} tolerance; complement norm, projection, output and null-membership checks have maximum scaled defect 5.10×10^{-16}. These consistency checks do not certify weak-mode forward accuracy or exact rank. Appendix E specifies their denominators.

The checkpoint-conditional θ-centred comparison uses the same fixed-scale covariance; Appendix E records its separate 95 checks.

Regional modal sums and conditional limits. Let $\lambda_i = s_i^2$, $q_{r,i} = \langle (U^\top Q_1^\top g(x))_i^2 \rangle_r$, $d_r = \langle \|Q_c^\top g(x)\|^2 \rangle_r$, and $A_r = d_r + \sum_i q_{r,i}$. Equation (3a) becomes

$$H_r(\gamma) = \frac{d_r}{\gamma} + \sum_i \frac{q_{r,i}}{\lambda_i + \gamma}, \qquad \langle \sigma^2 \rangle_r = v H_r(\gamma), \qquad R_\sigma^{\mathrm{rms}}(\gamma)^2 = \frac{H_{\mathcal{S}}(\gamma)}{H_{\mathcal{B}}(\gamma)}. \tag{9}$$

The conditional limits in Section 3.3 follow by multiplying (9) by γ: the strong limit requires $A_{\mathcal{B}} > 0$, and the finite weak ratio requires positive bulk energy in the whole kernel. The QR-complement ratio in Table 7 equals that weak limit if J has full row rank; tiny positive returned singular values do not certify this premise.

Neither limit bounds the intermediate quotient in general. Of the 17 full-time sampled curves, 13 have increasing and decreasing segments, and 12 leave the interval between the reported endpoint ratios. Allen–Cahn MLP has tangent/QR ratios about 1.27–2.53 but sampled maxima 7.30–7.73.

family	strong-γ ratio	QR-complement ratio	sampled R_σ^{rms}
Burgers–MLP	[0.9467, 1.023]	[42.71, 509.9]	[2.136, 517.3]
Burgers–Fourier	[1, 1.027]	[11.81, 30.54]	[1.361, 31.56]
Allen–Cahn–MLP	[1.457, 1.474]	[1.268, 2.532]	[1.465, 7.731]
Allen–Cahn–Fourier	[1.089, 1.138]	[25.21, 35.47]	[2.822, 31.08]

Table 7: Full-time regional RMS tangent-norm ratios, QR-complement ratios and measured contrast ranges across every saved γ and admitted run in each family. The QR column equals the whole-kernel weak limit when J has full row rank. Endpoint ratios are not global contrast bounds; the final column is a finite sampled range shared by all predictive centres.

The largest saved $\gamma/\lambda_{\max}$ is only 6.28×10^{-5}–2.96×10^{-4}; the saved modal sums also permit evaluation at stronger precisions without new fields. At $\gamma = 10^3\lambda_{\max}$, all full and Burgers post-formation contrasts are within 1.94×10^{-4} relative to their tangent-norm limits. The first-order expansion (4) differs from the returned squared ratios by at most 3.43×10^{-7} relative to those squared ratios. Appendix D reports changed-scale brackets and loading ratios at $\epsilon = \gamma$; the brackets remain too broad to certify a realised-error deficit.

Evidence selection and mechanism. For the zero-mean prior's affine observation $y = J\theta - r$, set $z = V^\top y$ and $C_i = v + \lambda_i/\alpha$. The affine log evidence (MacKay, 1992) is

$$\ell(\alpha, v) = -\tfrac{1}{2}\sum_{i=1}^{N}[z_i^2/C_i + \log C_i + \log(2\pi)].$$

Fourteen original selections occur at $10^{-4}\widehat{v}$ and remain on the new $10^{-8}\widehat{v}$ boundary after extension; the three Allen–Cahn Fourier selections are interior and unchanged (Table 8). All selected alpha coordinates are interior. These zero-mean-prior selections are best tested grid pairs, not continuous optima.

For the θ-centred prior the same formula uses $z = -V^\top r$. On the extended grid, 14 selections lie at the lower noise boundary, two Allen–Cahn Fourier selections at the upper alpha boundary, and one is interior. Sharp 1.96-sigma coverage around its affine mean is below nominal in all 17 runs, ranging from 0.00275 to 0.90175.

Each case changes singular values by $\delta s = \varepsilon_{64}\max(N, P)s_{\max}$ or $10\delta s$, upwards, downwards or alternately, with zero clamping. It also changes z coherently by $(s' - s)\theta_{\mathrm{modes}}$, where $\theta_{\mathrm{modes}} = U^\top Q_1^\top \theta$. These are finite sensitivity cases, not certified error bounds. All Allen–Cahn MLP selections change in all six cases; their $1\times$ quadratic terms $\delta s^2/\gamma$ alone are 7.55, 488.55 and 228.65. Their extended selections are not treated as robust scale estimates.

At fixed alpha,

$$D_v = \frac{\partial \ell}{\partial \log v} = -\frac{1}{2}\sum_i \frac{v}{C_i}\left(1 - \frac{z_i^2}{C_i}\right). \tag{10}$$

At the fourteen extended boundary selections, the $\lambda_i/\gamma < 0.01$ group contains 603–4594 modes, with summed projected data energy only 0.000481–0.014164 times summed model variance (Figure 3 illustrates Burgers–MLP seed 2). Its negative slope contributions favour lower noise. Original/extended fixed-alpha slopes are -2633 to -2561 / -2379 to -2255 for Burgers–MLP, -481

family	runs	extended noise edge	selected $\gamma/\gamma_{\text{ext}}$
Burgers–MLP	8	8/8	$[0.562, 3.162]$
Burgers–Fourier	3	3/3	$[1, 1.778]$
Allen–Cahn–MLP	3	3/3	$[1.778, 3.16228 \times 10^5]$
Allen–Cahn–Fourier	3	0/3	1

Table 8: Zero-mean-prior evidence-grid outcomes and six-case sensitivity. The interval is perturbed selected effective precision divided by the extended selection, across each family's runs and cases. Perturbations change singular values and the affine observation coherently, keeping singular vectors fixed; fields at alternative selected scales use the original Jacobian. Appendix D gives full per-run values.

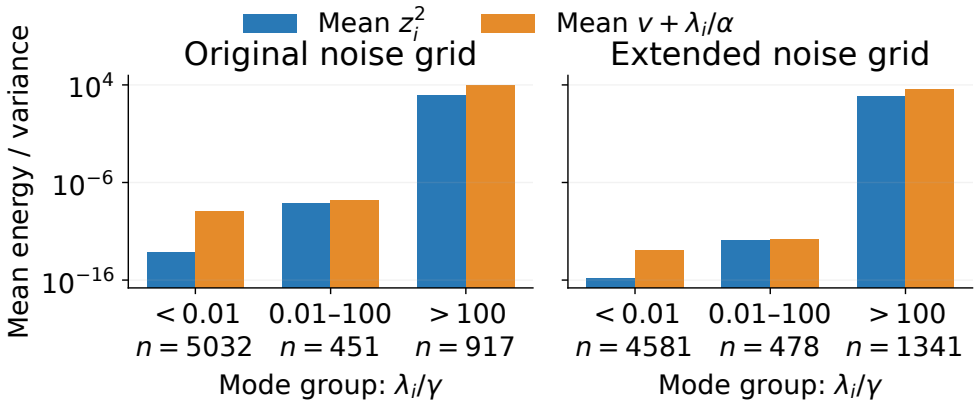


Figure 3: Burgers–MLP seed 2, zero-mean prior: projected affine-observation energy z_i^2 and model variance $v + \lambda_i/\alpha$ at the original and extended selections, averaged in three λ_i/γ groups. Counts accompany the bars. Little data energy in weak groups favours lower noise in (10); the grouping is descriptive.

to -417 / -407 to -317 for Burgers–Fourier, and -1503 to -1124 / -1397 to -753 for Allen–Cahn MLP. Allen–Cahn Fourier slopes are -291 to -113; an interior quarter-decade maximum need not have zero continuous slope. These fixed-alpha slopes describe local behaviour, not an alpha-reoptimised path or the limit $v \to 0$.

Synthetic validation. Dense tests give posterior mean/covariance relative discrepancies at most 4.90×10^{-12} and 3.01×10^{-12}, and log-evidence discrepancy 1.42×10^{-13}. Across 1000 correctly specified draws, posterior-centred coverage is 0.6870 (SE 0.00544) at one sigma and 0.9534 (SE 0.00249) at 1.96 sigma. A full-column-rank fit with $N = 40, P = 12, v = 0.2$ gives mean residual-noise MLE 0.140411 (SE 0.001198), consistent with $(N - P)v/N = 0.14$ rather than v. In each 100-draw evidence toy, including a 21.2-decade spectrum with $79/96$ eigenvalues below $\varepsilon_{64}\lambda_{\max}$, all selected pairs are interior and agree with the prescribed-spectrum reference. The stressed toy nevertheless fails its whole-grid 10^{-7} scaled tolerance, with maximum discrepancy 4.55×10^{-5},

			Baseline			Conditional MLE		
family	n	trained	zero	θ	nonlinear	zero	θ	nonlinear
Burgers–MLP	8	0.02546	$0.05883^{(7)}$	$0.02537^{(2)}$	$0.02537^{(2)}$	$0.023^{(5)}$	$0.02283^{(2)}$	$0.03944^{(6)}$
Burgers–Fourier	3	0.05124	$0.04316^{(1)}$	$0.05118^{(0)}$	$0.05118^{(0)}$	$0.04235^{(0)}$	$0.04935^{(0)}$	$0.05091^{(1)}$
Allen–Cahn–MLP	3	0.03287	$0.2776^{(3)}$	$0.03275^{(0)}$	$0.03275^{(0)}$	$0.02804^{(1)}$	$0.02455^{(0)}$	$0.2701^{(3)}$
Allen–Cahn–Fourier	3	0.03351	$0.245^{(3)}$	$0.03349^{(0)}$	$0.03349^{(0)}$	$0.03975^{(1)}$	$0.02496^{(0)}$	$0.03861^{(1)}$

Table 9: Paired full-grid RMS prediction errors. Entries are unweighted means of per-run RMS errors; superscripts count runs worse than the trained predictor out of n. Baseline uses $(\alpha_0, 1)$ and MLE uses $(\alpha_0, \widehat{v})$. Zero and θ denote the two affine prior-centre predictions; nonlinear is $u_{\theta+\delta_L}$, a forward diagnostic rather than a posterior mean. All comparisons retain the original covariance linearisation.

family	region	baseline	MLE
Burgers–MLP	bulk	$[7.54, 54.1]$	$[0.264, 1.51]$
	sharp	$[0.58, 57.7]$	$[0.108, 8.55]$
Burgers–Fourier	bulk	$[3.87, 8.31]$	$[0.823, 2.42]$
	sharp	$[0.666, 0.864]$	$[0.167, 0.809]$
Allen–Cahn–MLP	bulk	$[15.7, 23]$	$[1.57, 2.59]$
	sharp	$[5.46, 7.22]$	$[0.416, 0.628]$
Allen–Cahn–Fourier	bulk	$[8.94, 9.95]$	$[0.97, 1.09]$
	sharp	$[4.38, 6.25]$	$[0.36, 2.26]$

Table 10: Across-run ranges of total regional RMS zero-mean-prior affine displacement divided by trained error RMS on the same full-time region. Baseline is $(\alpha_0, 1)$ and MLE is $(\alpha_0, \widehat{v})$. Displacement alone does not determine prediction accuracy; its alignment with trained error matters. Appendix D includes transition regions and evidence settings.

despite selected log-evidence discrepancies at most 9.40×10^{-8}. Appendix E gives scale-estimation distributions and the failed check; these controls do not establish PINN evidence stability.

Centre and regional displacement. Table 9 compares the prior centres and nonlinear diagnostic at fixed baseline/MLE scales; Table 10 reports relative regional displacements. A reduced error contrast need not improve accuracy. Covariance is shared, whereas errors and coverage are centre-specific. At the conditional MLE, all eight Burgers–MLP zero-mean-prior affine gaps lie in $[1.51, 3.39]$; one-standard-deviation coverage is 0.9815–0.99998 in bulk and 0.4185–0.92475 in sharp. The trained-centre allocation reversals and bulk undercoverage therefore do not transfer to this centre.

Sampled oracle and post-formation coverage. At every saved γ, centre and subset, a higher empirical bulk-0.68 quantile of $|e_c|/\sigma$ supplies a width multiplier. Selection minimises $|\log \rho|$ across distinct saved precisions, with lowest γ breaking ties. This reference-error oracle uses sampled settings. Trained Burgers–MLP seeds $0, 4, 5, 6, 7$ have full-time sampled crossing brackets; six zero-mean affine full-time and six post-formation curves have brackets. Full-time zero-mean affine seeds $1, 7$ and post-formation seeds $1, 3$ stay above one at every saved point.

family	subset	trained 68%	affine 68%	trained 95%	affine 95%
Burgers–MLP	full	[0.3427, 0.8605]	[0.164, 0.9815]	[0.639, 0.9742]	[0.298, 0.995]
Burgers–MLP	post	[0.1639, 0.9959]	[0.008972, 0.3467]	[0.4176, 1]	[0.01387, 0.7121]
Burgers–Fourier	full	[0.1985, 0.5978]	[0.2828, 0.4662]	[0.4472, 0.7588]	[0.5145, 0.742]
Burgers–Fourier	post	[0.01713, 0.0938]	[0.08972, 0.3173]	[0.03589, 0.1819]	[0.1998, 0.5571]
Allen–Cahn–MLP	full	[0.0095, 0.0655]	[0.2015, 0.2755]	[0.0395, 0.2722]	[0.3588, 0.3962]
Allen–Cahn–Fourier	full	[0.1477, 0.2005]	[0.0775, 0.3827]	[0.2407, 0.4138]	[0.1432, 0.7338]

Table 11: Across-run sharp-region coverage ranges for trained and zero-mean-prior affine centres at each run's closest saved contrast setting, chosen separately for each centre and subset. A closest sample need not have $\rho \approx 1$, so these rows do not generally evaluate coverage after RMS-contrast matching. At each saved γ, a higher empirical bulk-0.68 quantile supplies the width multiplier; selection minimises $|\log \rho|$ and the lowest saved γ breaks exact ties. These reference-error oracles use all 8 Burgers–MLP or 3 other admitted runs, including runs without a crossing. "Post" uses the unchanged global masks intersected with Burgers $t \geq 1/\pi$. Bulk one-sigma coverage is about 0.68002; neither nominal sharp coverage nor an exact root is enforced.

For Burgers, post-formation masks intersect unchanged global masks with $t \geq 1/\pi$, yielding $48{,}246$ bulk, 4928 transition and 1226 sharp points. Table 11 includes all runs, including those without crossings; the closest sample need not have $\rho \approx 1$. Its coverage ranges distinguish regional contrast from calibration. Saved modal sums refine the five trained-centre crossings; direct root fields confirm their coverage envelopes (Table 12). These precisions are not contrast-matching roots for the other centres. Appendix D gives per-run curves.

Predictive bounds for the prior-centre comparison. For $\gamma, v > 0$, write $A_\gamma = F + \gamma I$. The zero-mean-prior shift splits as

$$\delta_\gamma = \delta_{L,\gamma} + \delta_{P,\gamma}, \qquad \delta_{L,\gamma} = -A_\gamma^{-1} J^\top r, \qquad \delta_{P,\gamma} = -\gamma A_\gamma^{-1} \theta.$$

In the factorisation above, their coordinates are $-s_i (V^\top r)_i/(s_i^2+\gamma)$ and $-\gamma\theta_{\mathrm{modes,i}}/(s_i^2+\gamma)$; the QR-complement contributions are 0 and $-Q_c Q_c^\top \theta$, respectively. The total norm includes their cross term. Replacing the prior by $\vartheta \sim \mathcal{N}(\theta, \alpha^{-1} I)$ removes $\delta_{P,\gamma}$ and leaves the covariance unchanged.

Define $\mathcal{K}_\gamma = r^\top J A_\gamma^{-1} J^\top r$ and $c_\gamma = \sqrt{\mathcal{K}_\gamma / v}$. Since $\delta_{L,\gamma}^\top A_\gamma \delta_{L,\gamma} = \mathcal{K}_\gamma$, Cauchy–Schwarz in the A_γ metric gives the pointwise bound

$$|m_\gamma^\theta(x) - u_\theta(x)| = |g(x)^\top \delta_{L,\gamma}| \leq \sigma(x) c_\gamma, \qquad \mathcal{K}_\gamma = \sum_i \frac{s_i^2 (V^\top r)_i^2}{s_i^2 + \gamma}. \tag{11}$$

Let E_r and Σ_r denote trained error RMS and uncertainty RMS, and let $q_{r,i}$ be the regional mode energies in (9). The regional RMS triangle inequality also gives

$$B_r = \min\left\{ c_\gamma \Sigma_r, \sum_i |\delta_{L,\gamma,i}| \sqrt{q_{r,i}} \right\}, \qquad \max(0, E_r - B_r) \leq E_r^\theta \leq E_r + B_r. \tag{12}$$

For $E_{\mathcal{B}}^\theta > 0$ and $R_\sigma^{\mathrm{rms}} > 0$, $\rho^\theta \geq \max(0, E_{\mathcal{S}} - B_{\mathcal{S}})/[(E_{\mathcal{B}} + B_{\mathcal{B}}) R_\sigma^{\mathrm{rms}}]$. The prior-only prediction differs from the archived zero-mean affine mean by $g^\top \delta_L$; its full-grid error is therefore at least

$E_{\text{full}}^{\text{zero}} - B_{\text{full}}$. This attributes worsening through a prediction-level bound rather than parameter norms. If $C_c(t)$ denotes empirical coverage at t standard deviations, then

$$C_{\text{trained}}(t - c_\gamma) \le C_\theta(t) \le C_{\text{trained}}(t + c_\gamma), \qquad t \ge c_\gamma,$$

by the pointwise triangle inequality.

The saved-array calculation reproduces all 401 total displacement norms to 4.44×10^{-16} relative and regional variances to 6.73×10^{-14}. Across those settings, $\|\delta_\gamma\|/\|\theta\|$ ranges from approximately 0.9889 to 31.17; these displacements do not validate a nonlinear first-order approximation. At the baseline, $\|\delta\|/\|\theta\|$ is $[0.997775, 0.999576]$, whereas $\|\delta_L\|/\|\theta\|$ is $[8.73 \times 10^{-6}, 1.5413 \times 10^{-4}]$; the QR-complement norm alone is $[0.976675, 0.985087]\|\theta\|$. Using (11) and (12) gives $c_\gamma \in [0.007489, 0.090775]$ and, after the regional refinement, establishes a θ-centred allocation deficit in $16/17$ runs from bounds alone. Its seed-4 lower bound 0.4878 is inconclusive; direct predictions establish the deficit in all 17 (Section 3.1). The prior-only prediction is worse in all 14 baseline cases where the zero-mean affine prediction worsens. The θ-centred one-sigma coverage lower bounds are 0.7925 in sharp and 0.997196 in bulk, so both regions overcover in all 17 runs. Bulk 1.96-sigma coverage is 1; sharp 1.96-sigma coverage is above nominal in 12 runs and below nominal in the other 5. Their upper bounds are 0.924, 0.883, 0.90875, 0.91225 and 0.896 for Burgers–MLP seeds $1, 2, 3$ and Burgers–Fourier seeds $0, 6$, respectively. At the conditional MLE, $c_\gamma \in [75.428, 79.910]$ is uninformative; Table 9 instead reports the directly evaluated alternative centre.

Coverage envelopes at refined contrast-matching roots. At unit noise, each pointwise variance has the positive modal representation $q_\gamma(x) = \sum_i a_i(x)/(\lambda_i + \gamma)$, including the QR complement as a zero-eigenvalue mode, with $a_i(x), \lambda_i \ge 0$. For positive endpoint variances at saved precisions $0 < l \le \gamma \le h$, $l < h$, and $t = (\gamma - l)/(h - l)$, positivity gives

$$L_\gamma = \frac{(1-t)lq_l + thq_h}{\gamma} \le q_\gamma \le U_\gamma = \left(\frac{1-t}{q_l} + \frac{t}{q_h}\right)^{-1}. \tag{13}$$

For the lower bound in (13), γq_γ is concave because every $\gamma/(\lambda_i + \gamma)$ is concave. For the upper bound, $q' = -\sum_i a_i/(\lambda_i + \gamma)^2$ and $q'' = 2\sum_i a_i/(\lambda_i + \gamma)^3$ satisfy $(q')^2 \le qq''/2$ by Cauchy–Schwarz. Hence $(1/q)'' = [2(q')^2 - qq'']/q^3 \le 0$, and taking the reciprocal of its chord gives U_γ. Both envelopes decrease with γ since $q_l \ge q_h$.

For a numerically refined root bracket $[a, b] \subset [l, h]$, use L_b and U_a to retain root-location uncertainty. The higher empirical bulk-0.68 quantile $Q_{\mathcal{B}}$ is monotone, so

$$k_{\min} = Q_{\mathcal{B}}(|e|/\sqrt{U_a}), \qquad k_{\max} = Q_{\mathcal{B}}(|e|/\sqrt{L_b})$$

enclose the multiplier k_γ that calibrates bulk coverage at the root. For sharp coverage at threshold $z \ge 0$, this yields

$$\left\langle \mathbf{1}\{|e| \le z k_{\min}\sqrt{L_b}\}\right\rangle_{\mathcal{S}} \le C_{\mathcal{S},\gamma}(z) \le \left\langle \mathbf{1}\{|e| \le z k_{\max}\sqrt{U_a}\}\right\rangle_{\mathcal{S}}. \tag{14}$$

These analytic bounds use the returned spectral model and saved endpoint fields; they are not directed-rounding certificates for the original Jacobian. Direct pointwise root evaluations confirm the enclosures in Table 12. Root refinement concerns the five observed trained-centre brackets, without asserting uniqueness or global optimality. Appendix F maps the Lean verification of these analytic bounds. The evaluated run counts and refined brackets remain numerical results.

		Sharp 1σ coverage		Sharp 1.96σ coverage	
seed	refined γ	direct	envelope	direct	envelope
0	0.02105515	0.3275	$[0.323, 0.333]$	0.78275	$[0.7815, 0.78475]$
4	0.73626974	0.427	$[0.41975, 0.43375]$	0.852	$[0.84525, 0.862]$
5	2.0476385×10^{-5}	0.30325	$[0.29775, 0.315]$	0.67725	$[0.6755, 0.67925]$
6	0.00163541	0.52325	$[0.5225, 0.526]$	0.847	$[0.8465, 0.84775]$
7	3.0590419×10^{-5}	0.55725	$[0.554, 0.563]$	0.8235	$[0.82125, 0.82675]$

Table 12: Direct sharp coverage and saved envelopes at five trained Burgers–MLP roots. Displayed precisions are rounded; root brackets have relative width below 5.3×10^{-13}. Bulk calibration uses the higher empirical 0.68 quantile. Envelopes include root and calibration uncertainty through (14); all direct values lie inside.

Regional calibration and sampled agreement band. At each trained-centre closest saved setting, use either $k_{\mathcal{B}} = Q_{\mathcal{B}}(|e|/\sigma; 0.68)$ or $k_{\mathcal{S}} = Q_{\mathcal{S}}(|e|/\sigma; 0.68)$, with the higher empirical quantile. Write $C^r_{\mathcal{S}}(z) = \langle \mathbf{1}\{|e| \le z k_r \sigma\}\rangle_{\mathcal{S}}$ and $Z^r = e/(k_r\sigma)$ for the resulting coverage and standardised errors. Table 13 gives full-grid per-seed results and family ranges, including the Burgers post-formation subsets. Sharp calibration fixes its one-standard-deviation coverage by construction: 0.68025 on 4000 full-grid sharp points and 0.68108 on 1226 post-formation points here. The informative outputs are coverage at 1.96 standard deviations and the absolute and signed standardised means. These are finite-grid descriptions, not repeated-draw guarantees. Inclusive quantile comparisons can differ by one point from older multiplied-width arithmetic; all continuous summaries reproduce to relative tolerance 10^{-12}.

The band inventory retains every distinct saved precision with $0.9 \le \rho \le 1.1$: 28 run/subset settings across the full grids and Burgers post-formation masks. For Burgers–MLP seed 7, the seven full-grid samples give sharp coverage 0.44725–0.986 at one standard deviation and 0.78375–1 at 1.96 after bulk calibration. Their ρ values span 0.90635–1.06370; nearby contrast values need not correspond to nearby precisions on a nonmonotone curve. The other four crossing seeds each have one saved full-grid point in the band, with one-standard-deviation coverages 0.34275, 0.437, 0.35925 and 0.5165 for seeds $0, 4, 5, 6$. The archived per-run inventory is sampled rather than a continuous-band enclosure; Table 12 retains the separate refined-root bounds.

Width versus allocation. At $v = 1$, changing γ changes modal weights. At fixed $\gamma = \alpha_0$, varying v with $\alpha = \alpha_0/v$ multiplies widths by $\sqrt{v}$ without changing contrast: $\overline{Z}$ scales as $v^{-1/2}$ and coverage is the empirical CDF of fixed standardised errors at $\sqrt{v}$. Figure 4 shows both effects; the fixed-γ curves are rescalings of saved fields, not fitted scales.

Appendix D. Extended Empirical Records

The extended records use the fixed 17-checkpoint cohort, global reference masks and raw weighted residuals (Definitions 1 and 2). Covariance contrast is shared by the predictive centres at fixed scales; error contrast and the gap are centre-specific (Definition 3). "Full" denotes the complete evaluation grid. Burgers "post" intersects the unchanged masks with $t \ge 1/\pi$; no Allen–Cahn post-formation subset is imposed. Figures 5 and 6 show covariance contrasts and trained/zero-mean affine gaps; Table 9 adds the θ-centred comparison.

Burgers–MLP seed	ρ	$C_{\mathcal{S}}^{\mathcal{B}}(1)$	$C_{\mathcal{S}}^{\mathcal{B}}(1.96)$	$C_{\mathcal{S}}^{\mathcal{S}}(1.96)$	$\overline{\lvert Z\rvert}^{\mathcal{S}}$	$\overline{Z}^{\mathcal{S}}$
0	0.95438	0.34275	0.7885	0.92025	0.91339	−0.24502
1	1.8363	0.8605	0.97425	0.865	0.94283	0.85765
2	2.5874	0.44675	0.639	0.85025	1.3644	1.209
3	1.9924	0.725	0.933	0.9175	0.77179	0.56348
4	0.95486	0.437	0.86375	0.95775	0.83109	−0.069397
5	0.97331	0.35925	0.69175	0.8345	1.0436	−1.0371
6	1.0323	0.5165	0.8435	0.92825	0.85021	0.069544
7	0.96422	0.65175	0.837	0.84675	1.034	0.79421

Family / subset (n)	$C_{\mathcal{S}}^{\mathcal{S}}(1.96)$	$\overline{\lvert Z\rvert}^{\mathcal{S}}$	$\overline{Z}^{\mathcal{S}}$
Burgers–MLP / full (8)	[0.8345, 0.95775]	[0.77179, 1.3644]	[−1.0371, 1.209]
Burgers–Fourier / full (3)	[0.82475, 0.85075]	[1.0792, 1.3448]	[0.98518, 1.3234]
Allen–Cahn–MLP / full (3)	[0.921, 0.959]	[0.77708, 0.86039]	[−0.29925, 0.32371]
Allen–Cahn–Fourier / full (3)	[0.98325, 0.9995]	[0.71247, 0.81491]	[−0.36263, 0.27285]
Burgers–MLP / post (8)	[0.94454, 1]	[0.61868, 0.98575]	[−0.87154, 0.82267]
Burgers–Fourier / post (3)	[0.8646, 1]	[0.82852, 0.87527]	[0.7548, 0.86578]

Table 13: Trained-centre regional calibration at each run's closest saved contrast setting. The superscript $\mathcal{B}$ or $\mathcal{S}$ denotes the region whose higher empirical 0.68 quantile supplies the width multiplier. The upper panel gives eight full-grid Burgers–MLP seeds; the lower panel gives observed ranges for all 17 full-grid runs and the Burgers post-formation subsets. Each setting minimises $|\log \rho|$, with ties at the lowest saved precision. $\overline{\lvert Z\rvert}^{\mathcal{S}}$ and $\overline{Z}^{\mathcal{S}}$ are absolute and signed means of sharp-calibrated standardised errors. Sharp one-standard-deviation coverage is fixed by construction, up to discreteness and ties. Displayed values are rounded; reference calibration does not define a training rule.

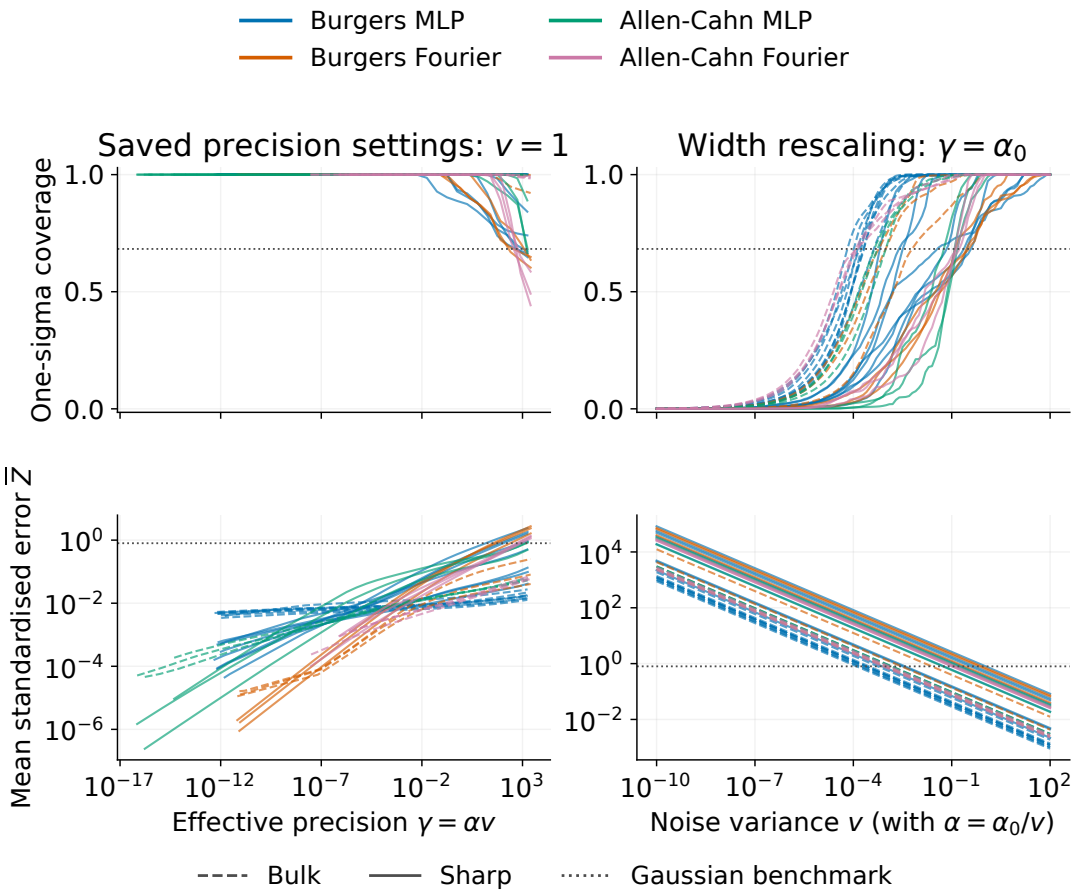


Figure 4: Trained-centre one-sigma coverage and mean standardised error for all 17 full-time runs. Colours identify families; dashed/solid curves denote bulk/sharp. Left: saved precisions at $v = 1$; right: width rescaling at $\gamma = \alpha_0$. Dotted lines mark Gaussian expectations 0.68269 and 0.79788. Lines join evaluated points without asserting confidence intervals.

D.1 Evidence Selections and Regional Spectral Summaries

Table 14 extends the zero-mean-prior evidence summary in Appendix C. The initial evidence grid has lower noise edge $10^{-4}\widehat{v}$ and its one extension has lower edge $10^{-8}\widehat{v}$. The six perturbation cases are specified in Appendix E; selections are finite-grid candidates and displayed ranges are descriptive.

Changed-scale spectral partition. Let $q_{r,i} = \langle (U^\top Q_1^\top g)_i^2 \rangle_r$ and $d_r = \langle \|Q_c^\top g\|^2 \rangle_r$, using the factorisation in Appendix C. At $\epsilon = \gamma$, these saved energies give $Q_{r,\mathrm{sl}} = d_r + \sum_{\lambda_i \le \gamma} q_{r,i}$ and $Q_{r,\mathrm{st}} = \sum_{\lambda_i > \gamma} q_{r,i}$. Substitution into Equation (5), with prior precision replaced by γ, gives the returned-spectrum brackets in Table 15. Table 16 gives unweighted energy ratios. Neither certifies weak-mode forward accuracy.

run	a_{orig}	a_{ext}	$v_{\mathrm{ext}}/\widehat{v}$	γ_{ext}	perturbed $\gamma/\gamma_{\mathrm{ext}}$	$\delta s^2/\gamma_{\mathrm{ext}}$
BM0	0.316	0.316	1×10^{-8}	6.64×10^{-13}	[1, 3.16]	0.0619
BM1	0.562	0.562	1×10^{-8}	1.59×10^{-12}	[1, 1.78]	0.0338
BM2	0.316	0.562	1×10^{-8}	1.24×10^{-12}	[0.562, 1.78]	0.0289
BM3	0.562	0.562	1×10^{-8}	7.87×10^{-13}	[1, 1.78]	0.0408
BM4	0.316	0.316	1×10^{-8}	4.95×10^{-13}	[1, 3.16]	0.0832
BM5	0.562	0.562	1×10^{-8}	1.06×10^{-12}	[1, 1.78]	0.0256
BM6	0.562	0.562	1×10^{-8}	6.99×10^{-13}	[1, 1.78]	0.0555
BM7	0.562	0.562	1×10^{-8}	1.03×10^{-12}	[1, 1.78]	0.0297
BF0	1	1	1×10^{-8}	6.71×10^{-12}	[1, 1]	0.00666
BF2	1	1	1×10^{-8}	8.52×10^{-12}	[1, 1.78]	0.00844
BF6	1	1	1×10^{-8}	9.23×10^{-12}	[1, 1]	0.00665
AM0	0.01	5.62×10^{-4}	1×10^{-8}	4.88×10^{-15}	[1.78, 1000]	7.55
AM1	0.00316	1.78×10^{-5}	1×10^{-8}	7.07×10^{-17}	[17.8, 3.16×10^{5}]	489
AM3	0.00562	3.16×10^{-5}	1×10^{-8}	1.69×10^{-16}	[5.62, 3.16×10^{5}]	229
AF0	0.562	0.562	0.01	8.03×10^{-7}	[1, 1]	4.83×10^{-8}
AF1	0.316	0.316	0.00178	3.11×10^{-8}	[1, 1]	1.63×10^{-6}
AF3	0.562	0.562	0.0562	4.85×10^{-6}	[1, 1]	6.6×10^{-9}

Table 14: Zero-mean-prior evidence selections and six-case sensitivity. BM/BF and AM/AF denote Burgers and Allen–Cahn MLP/Fourier, with seed suffixes; $a = \alpha/\alpha_0$. BM/BF/AM select the lower noise edges specified above; AF selections are interior and unchanged. The $1\times$ envelope δs gives the quadratic term, not a uniform eigenvalue error bound.

family	conditional MLE	extended evidence
Burgers–MLP	$[0.00491,\ 7.01\times10^{4}]$	$[0.00237,\ 1.88\times10^{5}]$
Burgers–Fourier	$[0.0258,\ 1.13\times10^{3}]$	$[0.0222,\ 1.43\times10^{3}]$
Allen–Cahn–MLP	[0.0079, 713]	$[0.00105,\ 3.06\times10^{3}]$
Allen–Cahn–Fourier	$[0.0161,\ 1.15\times10^{3}]$	$[0.0124,\ 1.97\times10^{3}]$

Table 15: Changed-scale two-block brackets at $\epsilon = \gamma$ from saved direct-J regional energies, in σ-ratio units. Intervals enclose the union of run-specific returned-spectrum brackets, rounded outward. Weak-mode forward accuracy remains uncertified, and the broad brackets cannot establish the realised-error deficit.

D.2 Displacement under the Zero-Mean Prior

Table 17 extends Table 10 to transition regions and evidence settings. For the shift $c + b_\gamma$, the scale-independent component $c(x) = -g(x)^\top Q_c Q_c^\top \theta$ gives $\langle (c+b_\gamma)^2\rangle_r = \langle c^2\rangle_r + \langle b_\gamma^2\rangle_r + 2\langle c b_\gamma\rangle_r$. Every full-grid sharp complement field contains both signs. For Burgers–MLP, its RMS is 0.001999–0.06048 and signed mean -0.02154–0.004895; $4/8$ baseline sharp cross terms are destructive. For Burgers–Fourier seed 0, complement sharp RMS 0.04236 exceeds total MLE shift RMS 0.04102. Component RMS values are nonadditive.

family	setting	$Q_{\mathcal{S},\mathrm{sl}}/Q_{\mathcal{B},\mathrm{sl}}$	$Q_{\mathcal{S},\mathrm{st}}/Q_{\mathcal{B},\mathrm{st}}$
Burgers–MLP	MLE	$[2.15 \times 10^3,\ 1.94 \times 10^5]$	[0.896, 1.05]
	original evidence	$[1.95 \times 10^3,\ 2.67 \times 10^5]$	[0.896, 1.05]
	extended evidence	$[1.91 \times 10^3,\ 2.67 \times 10^5]$	[0.896, 1.05]
Burgers–Fourier	MLE	[150, 832]	[0.999, 1.05]
	original evidence	[150, 998]	[0.999, 1.05]
	extended evidence	[145, 996]	[0.999, 1.05]
Allen–Cahn–MLP	MLE	[14, 20.5]	[2.12, 2.17]
	original evidence	[4.59, 9.76]	[2.12, 2.17]
	extended evidence	[2.14, 7.02]	[2.12, 2.17]
Allen–Cahn–Fourier	MLE	[291, 693]	[1.19, 1.29]
	original evidence	[476, 988]	[1.19, 1.29]
	extended evidence	[476, 988]	[1.19, 1.29]

Table 16: Changed-scale sharp-to-bulk mean-square tangent loading ratios on the full-time regions. Entries give across-run ranges. At each saved setting, weak loading sums returned modes with $\lambda_i \leq \gamma$ and the QR complement; stiff loading sums the remaining modes. Thus the partition uses $\epsilon = \gamma$, as in Table 15. These are energy ratios, without a square root or covariance weights, and are not uncertainty contrasts. The returned weak-mode split does not certify exact rank or weak-mode forward accuracy. Evidence settings are best tested zero-mean-prior grid selections.

family	region	baseline	MLE	original evidence	extended evidence
Burgers–MLP	bulk	[7.54, 54.1]	[0.264, 1.51]	[0.338, 1.53]	[0.349, 1.49]
	transition	[9.07, 109]	[0.288, 1.69]	[0.266, 1.53]	[0.286, 1.42]
	sharp	[0.58, 57.7]	[0.108, 8.55]	[0.114, 14.9]	[0.122, 16.9]
Burgers–Fourier	bulk	[3.87, 8.31]	[0.823, 2.42]	[0.714, 1.97]	[0.679, 1.84]
	transition	[3.51, 7.08]	[0.955, 1.8]	[0.807, 1.53]	[0.804, 1.42]
	sharp	[0.666, 0.864]	[0.167, 0.809]	[0.156, 0.695]	[0.155, 0.762]
Allen–Cahn–MLP	bulk	[15.7, 23]	[1.57, 2.59]	[2.43, 10.9]	[21.8, 30.5]
	transition	[10.8, 13.9]	[0.868, 1.64]	[1.18, 3.32]	[4.05, 21.2]
	sharp	[5.46, 7.22]	[0.416, 0.628]	[0.626, 0.81]	[0.626, 5.07]
Allen–Cahn–Fourier	bulk	[8.94, 9.95]	[0.97, 1.09]	[0.748, 0.963]	[0.748, 0.963]
	transition	[9.86, 11]	[0.794, 1.74]	[0.631, 1.13]	[0.631, 1.13]
	sharp	[4.38, 6.25]	[0.36, 2.26]	[0.331, 1.45]	[0.331, 1.45]

Table 17: Regional RMS zero-mean-prior affine posterior-mean shift divided by trained-prediction RMS error, with both quantities evaluated on the same full-time region. Entries give across-run ranges (8 Burgers–MLP runs and 3 in each other family). Baseline is $(\alpha_0, 1)$; MLE is $(\alpha_0, \widehat{v})$. Original and extended evidence columns use best tested grid settings, not certified continuous optima. A ratio above one means that the displacement exceeds the trained error in regional RMS; it does not determine whether the displaced prediction is more accurate.

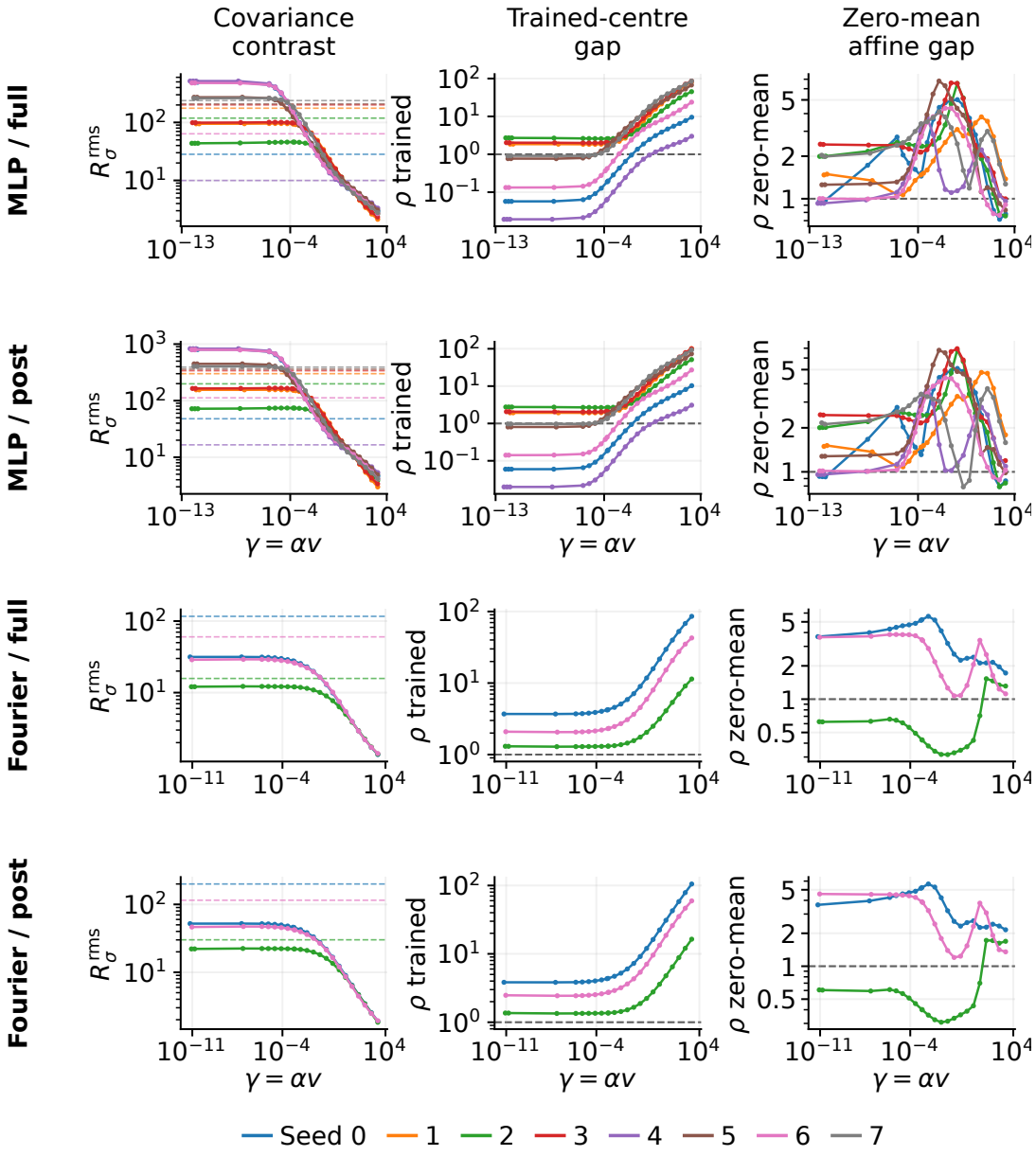


Figure 5: Burgers: eight admitted MLP and three admitted Fourier runs. Rows show MLP full-grid/post-formation, then Fourier full-grid/post-formation; post-formation intersects unchanged global masks with $t \geq 1/\pi$. Columns show RMS covariance contrast, trained gap and zero-mean affine gap; each gap uses its centre's error target. Coloured dashed lines give trained error targets; grey lines mark $\rho = 1$. Every distinct saved γ, including extensions and sensitivity aliases, uses the original Jacobian. Lines join samples without establishing monotonicity or a weak limit.

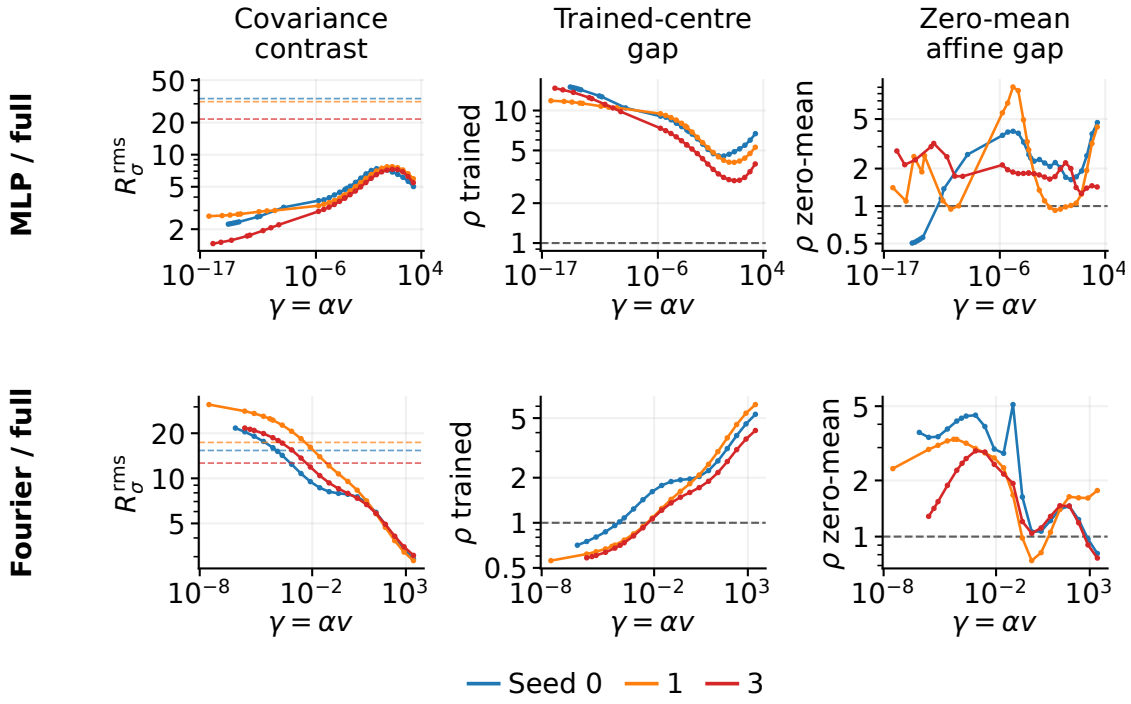


Figure 6: Allen–Cahn full-grid scale and centre curves for three admitted MLP (top) and three admitted Fourier (bottom) checkpoints. Columns and dashed lines follow Figure 5; each gap uses its centre's error target. Evidence settings for both families are in Table 14. All panels use the original Jacobian; lines connect saved settings.

Appendix E. Computation and Numerical Validation

This record details the numerical checks in Appendix C for the working covariance, prior-centre comparison and nonlinear forward diagnostic of Section 2.

E.1 Direct-J Evaluation and Consistency Checks

Appendix C specifies the direct-J factorisation and scale grids. Within each prior-centre model, shared γ settings have identical affine means and unit-variance fields; widths scale by $\sqrt{v}$.

At all 401 zero-mean-prior settings, automatic differentiation of the frozen residual map checks the normal-equation residual independently of the QR/SVD factors:

$$h_\gamma = J^\top(r + J\delta_\gamma) + \gamma(\theta + \delta_\gamma),$$
$$\eta_{\mathrm{back}} = \frac{\|h_\gamma\|}{(\lambda_{\max} + \gamma)\|\delta_\gamma\| + \|J^\top r + \gamma\theta\|}.$$

All 401 ratios are below the prescribed 10^{-7} tolerance, with maximum 6.19×10^{-17}; absolute residuals, denominators and shift norms are recorded. The complement shift $\delta_c = -Q_c Q_c^\top \theta$ is checked

through $J\delta_c$, its norm, projection identity and output. All 17 checkpoints pass, with maximum scaled defect 5.10×10^{-16}. Normwise residuals can miss weak-direction errors; null membership alone does not establish amplitude or completeness.

The θ-centred checks cover 95 distinct checkpoint–precision pairs: baseline, conditional MLE, original and extended evidence selections under both priors, and five trained-centre roots, merging repeated precisions per checkpoint. For these checks, use $h_\gamma^\theta = J^\top(r+J\delta_L)+\gamma\delta_L$ and denominator $(\lambda_{\max}+\gamma)\|\delta_L\|+\|J^\top r\|$; the maximum ratio is 9.12×10^{-17}. Nonlinear evaluation at $\theta+\delta_L$ leaves J, g and covariance fixed. Independent NumPy predictions at nine points per setting agree within 4.72×10^{-12} absolute. This is a deterministic forward check, not posterior sampling.

Fields reproduce regional metrics, and spectral sums reproduce regional mean variances. All 34 zero-mean-prior evidence selections and one-log-unit sets reproduce from saved surfaces; all 34 θ-centred selections reproduce from new factors.

Saved arrays reproduce all 401 displacement norms and regional variances (Appendix C). The envelopes enclose every tested stored intermediate variance field and a prescribed-spectrum example with a null mode at 49 interior precisions. Direct fields at the five refined trained-centre roots also lie within their coverage envelopes. These checks do not certify the original Jacobian by interval arithmetic.

E.2 Evidence Sensitivity Protocol

The six zero-mean-prior sensitivity cases in Appendix C keep singular vectors fixed and adjust the observation coherently: $z' = z + (s' - s)\theta_{\text{modes}}$, with $z = V^\top(J\theta - r)$ and $\theta_{\text{modes}} = U^\top Q_1^\top \theta$. For $\gamma = \alpha v$, an assumed singular-value budget gives

$$|\Delta\lambda_i| \le 2s_i|\Delta s_i| + |\Delta s_i|^2.$$

At $s_i = \sqrt{\gamma}$, the relative envelope is $2\delta s/\sqrt{\gamma} + \delta s^2/\gamma$; the $10\times$ case scales its two terms by 10 and 100. Table 14 shows why the extended Allen–Cahn MLP selections are especially exposed: the $1\times$ quadratic terms alone are 7.55, 488.55 and 228.65 times γ. The formed-Gram roundoff scale in Table 5 is not an error bound for direct-J QR/SVD.

Every Allen–Cahn Fourier selected pair and the pairs for Burgers–Fourier seeds $0, 6$ are unchanged in all six cases. Burgers–Fourier seed 2 moves its noise and γ up one quarter decade in the increase-$10\times$ case. Every Burgers–MLP run changes the selected pair in at least two cases, retaining its noise boundary; alpha ratios relative to its extended selection range from 0.562 to 3.162 across the cohort. Allen–Cahn MLP changes in all six cases per run: perturbed noise ratios reach 177.8 (seed 0) and 3162 (seeds $1, 3$), while γ ratios reach 1000 and 316,228, respectively. Changes in log evidence evaluated at the fixed extended pair reach about -1.05×10^6. Its extended scales therefore are not presented as robust scale estimates. Fields at the alternative selected scales use the original Jacobian, without recomputing a perturbed posterior.

For the fixed-alpha derivative D_v in (10), the signed terms are grouped by $\lambda_i/\gamma < 0.01$, 0.01–100, and > 100. Its weak-group energy ratios and counts are reported in Appendix C. For a cluster with all $\lambda_i \ll \gamma$, the leading contribution is $-\frac{1}{2}[m\log v + \sum_i z_i^2/v]$ apart from the constant; calculations retain the full formula. At fixed α and for this isolated cluster, the stationary variance is $v_* \approx m^{-1}\sum_i z_i^2$, provided all modes remain weak at v_*. For an exact-null cluster with positive projected energy this expression is exact. Decreasing v can invalidate $\lambda_i \ll \alpha v$ for positive eigenvalues; other modes and changed cluster membership affect the full optimum. The expression

therefore explains a local boundary tendency without settling the $v \to 0$ limit. Grouped energies avoid interpreting arbitrary individual coordinates. Allen–Cahn Fourier has substantial positive transition-group contributions, partly offsetting its negative weak group. The ratios are descriptive, not goodness-of-fit tests.

The fixed-alpha slope ranges are in Appendix C; they do not describe an alpha-reoptimised path or settle the limit $v \to 0$. The largest slope discrepancy against a local central log-noise finite difference is 4.43×10^{-6} absolute, or 5.51×10^{-9} after division by $1 + |D_v|$. Modal and grouped sums verify the calculation, not evidence-model adequacy.

E.3 Synthetic Controls

The small 12×20 design compares QR/SVD posterior means, covariances and evidence with dense calculations, with maximum discrepancies 4.90×10^{-12} and 3.01×10^{-12} in relative norm and 1.42×10^{-13} in absolute log evidence. In 1000 correctly specified prior/noise draws, the affine posterior output mean gives one-sigma coverage 0.6870 (draw-level SE 0.00544) and 95% coverage 0.9534 (SE 0.00249). A separate full-column-rank least-squares example with $N = 40, P = 12$ and generating noise variance $v = 0.2$ gives a mean of 0.140411 (SE 0.001198) for the fitted-residual estimator $\|r_{\mathrm{fit}}\|^2/N$, consistent with $(N - P)v/N = 0.14$. In a general linear fit the factor is $(N - \mathrm{rank}(X))/N$, reducing to $(N - P)/N$ only for full column rank. For a full-row-rank interpolating design it is zero; $P > N$ alone neither establishes that rank nor supplies a correction for nonlinear PINN fitting. Fitting can therefore reduce this estimator even under a correct noise model; this example is not an estimate of PINN discrepancy.

The fixed 100-draw overparameterised 16-decade eigenvalue toy has $100/100$ interior selections agreeing with its prescribed-spectrum reference. At the $5, 25, 50, 75, 95$ percentiles, the selected log-scale ratios $\log_{10}(\widehat{\alpha}/\alpha)$ are $(0, 0, 0, 0, 0.25)$; $\log_{10}(\widehat{v}/v)$ and $\log_{10}(\widehat{\gamma}/\gamma)$ are $(-0.25, 0, 0, 0, 0.25)$. These quarter-decade-grid medians coincide with the generating pair without establishing exact scale recovery.

A 21.2-decade synthetic spectrum with $N = 96, P = 160$ has $79/96$ prescribed eigenvalues below $\varepsilon_{64}\lambda_{\max}$. Its generating $\gamma = 10^{-8}$ is interior to the search grid. All 100 selected pairs are interior and agree with the prescribed-spectrum reference. The same log-scale percentiles are $(-0.25, 0, 0, 0, 0.25)$ for α and γ, and $(0, 0, 0, 0, 0)$ for v. The maximum selected log-evidence discrepancy is 9.40×10^{-8}. Across the whole tested grid, however, the maximum absolute discrepancy is 0.01775 and the maximum scaled discrepancy $|\ell_{\mathrm{num}} - \ell_{\mathrm{ref}}|/(1 + |\ell_{\mathrm{ref}}|)$ is 4.55×10^{-5}, exceeding the prescribed 10^{-7} tolerance. Here ℓ_{num} and ℓ_{ref} include the Gaussian normalising constant. Near-best discrepancies are small. Cancellation and the additive constant affect this scaled error; the failed whole-grid check alone establishes neither an incorrect selector nor model inadequacy. These controls retain the original tolerance and do not establish PINN evidence stability.

Appendix F. Formal Development

The independent `PaperReliability` development uses Lean `v4.30.0` (de Moura and Ullrich, 2021) and the matching Mathlib release (The mathlib Community, 2020). Table 18 records its analytic scope. The covariance bridge assumes a supplied exact orthogonal spectral representation. Physical modal applications use nonnegative eigenvalues and energies, positive precision and the stated positive pointwise or bulk denominators. Kernel statements concern exact zero modes, including the whole kernel rather than a numerical rank cutoff. Coverage envelopes use fixed trained

errors and regions, an actual empirical higher quantile, and a precision supplied within the retained bracket.

manuscript result	Lean declarations
Matrix-to-modal covariance, (3)	`spectral_covariance_representation`
Fixed-cutoff contrast bounds, (7)	`two_subspace_contrast_enclosure`
Regional variance aggregation, (6)	`regional_two_subspace_enclosure`
Sharpness witnesses, Proposition 9	`upper_envelope_arbitrary_energy_sharpness` `lower_envelope_arbitrary_energy_sharpness`
Conditional modal limits, Section 3.3	`regional_contrast_strong_limit` `regional_contrast_weak_limit`
Nonmonotone counterexample, Section 3.3	`endpoint_limits_do_not_bound_contrast`
Exact-kernel prior attribution, Appendix C	`kernel_prediction_shift_attribution`
Pointwise likelihood-shift bound, (11)	`likelihood_predictive_shift_bound`
Regional prior-centre bounds, (12)	`likelihood_regional_error_bounds`
Prior-centre coverage brackets, Appendix C	`likelihood_centre_coverage`
Positive-modal and retained-bracket envelopes, (13)	`positive_modal_variance_enclosure` `variance_enclosure_on_precision_bracket`
Bulk-calibrated sharp coverage, (14)	`calibrated_modal_coverage_enclosure`

Table 18: The authoritative map to the 15 public analytic declarations, all in namespace `PaperReliability` and free of `sorry`. Private proof lemmas are supporting infrastructure.

Verification covers fixed-cutoff inequalities and sharpness witnesses with prescribed energies in spectra bounded by $\lambda_{\max}$ and explicit approach factors as $\epsilon' \downarrow \epsilon$. Maximum-eigenvalue equalities and supremum/infimum statements are not separate formal results. The expansion (4), secondary mean-width identities and transfers, Remarks 5–6, and spectral-decomposition existence remain unverified. The proofs concern real-model inequalities, not QR/SVD accuracy, saved values, run counts or numerical root brackets.

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